TWO RANK-ONE DEGENERACIES IN THE MOVING-SOFA SECOND VARIATION, AND A MOVING-BREAKPOINT OBSTRUCTION TO LOCAL MAXIMALITY

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ABSTRACT. We study the second variation of the moving-sofa area functional F at Joseph Gerver's 1992 candidate sofa c_G, on the Sobolev space $H^2([0,\pi/2];\mathbb{R}^2)$ of corner trajectories modulo the two-dimensional translation-symmetry quotient.

Our starting point is a structural obstruction that has, we believe, been overlooked. Writing the smooth per-arc second variation in the form $C(\theta)\langle\eta,\eta'\rangle + D(\theta)\langle\eta,\eta''\rangle$ with ψ the active normal angle (Lemma 8), the matrix $D = \frac{1}{2}\left(\begin{smallmatrix} 1+\cos\psi & \sin\psi \\ \sin\psi & 1-\cos\psi \end{smallmatrix}\right)$ satisfies $\operatorname{tr} D = 1$ and

$$\det D \;\equiv\; 0,$$

identically in ψ: it is the *rank-one* projection $D = \hat{n}\hat{n}^\top$ onto the active wall normal. A single contact therefore controls only $\langle\eta',\hat{n}\rangle$ and is blind to the tangential component of η', so no coercivity statement can rest on the one-contact form.

The resolution is that at each θ several walls are in contact simultaneously, and the correct principal symbol is the sum $D_{\mathrm{tot}} = n_\mu\,\mu_\theta\mu_\theta^\top + n_\nu\,\nu_\theta\nu_\theta^\top$ over Romik's contact paths, where A, B contribute the normal $\pm\mu_\theta$ and C, D contribute $\pm\nu_\theta$. Because $\mu_\theta \perp \nu_\theta$, the eigenvalues of D_{tot} are exactly the contact counts (n_μ, n_ν), which Romik's five-phase decomposition fixes as $(1,2), (1,2), (1,1), (2,1), (2,1)$. Hence

$$\lambda_{\min}\big(D_{\mathrm{tot}}(\theta)\big) \;=\; \min(n_\mu, n_\nu) \;=\; 1 \qquad \text{for every } \theta,$$

a uniform lower bound with no free constant. The resulting prediction $\langle e_x, D_{\mathrm{tot}} e_x\rangle = n_\mu \cos^2\theta + n_\nu \sin^2\theta$ agrees with directly measured Rayleigh quotients to three or four decimal places.

This yields a Gårding inequality $Q(\eta) \le -(1-\varepsilon)\|\eta'\|_{L^2}^2 + K\|\eta\|_{L^2}^2$, and since $H^1 \hookrightarrow\hookrightarrow L^2$, coercivity in H^1 modulo a finite-dimensional subspace; the computed 32×32 Hessian, whose generalised eigenvalues against the H^1 Gram matrix are all strictly negative ($m_1 = 0.596$ at $N = 16$), closes that subspace. Coercivity in H^2 is by contrast *impossible*: $Q[k,k] = O(k^2)$ while $\|\varphi_k\|_{H^2}^2 = O(k^4)$, so the H^2 Rayleigh quotient tends to 0 (Theorem 2).

A second rank-one degeneracy appears in the first-order coefficient: $C^\top C$ also has determinant zero, so $\|C\|_{\mathrm{op}} = 2$ exactly, and the Gårding inequality becomes fully explicit, $Q(\eta) \le -\frac{1}{2}\|\eta'\|_{L^2}^2 + 18\|\eta\|_{L^2}^2$.

The passage from this to a local maximum nevertheless fails, and we locate precisely where. Since each contact path is affine in (c, c'), the Green's-theorem integrand is a product of two affine functions and hence *quadratic* in the jet (c, c', c''): within a fixed combinatorial cell F is an exactly quadratic functional, and the entire third variation is carried by the four *moving* breakpoints $b_j(c)$. Among the monomials these produce, exactly one — $\eta'(b_j)^3$ — grows fast enough to defeat an H^1 estimate. Measured against the area oracle, its coefficient is *nonzero* (≈ 1.09 and ≈ 1.89 in the two components, with control points vanishing identically and the two interior breakpoints exactly antisymmetric). The improved-Taylor hypothesis of Ioffe [10] and Dambrine–Lamboley [11] therefore fails in the pair (H^1, H^2), and no strict-local-maximum conclusion follows. For c_G this costs nothing — local maximality is a consequence of Baek's [5] global resolution — but it means the second-variation route does not independently establish it.

For Romik's ambidextrous candidate Σ ($A_R^\star \approx 1.64495$) the obstruction is compounded: its multi-contact symbol is $\approx 4I$ on the middle phase but degenerates at the end caps like $\lambda_{\min} \approx 4.95\,\theta^{1.69}$, so no uniform Gårding constant exists there at all. Local maximality of Σ — the second half of Romik's Open Problem 1 — accordingly remains open, and we identify two distinct reasons why.

The finite-dimensional Hessian throughout is evaluated with a floating-point polygon-intersection oracle rather than interval arithmetic; this affects the numerical constants but none of the structural conclusions, which are symbolic. The supporting computations are: (i) a high-precision Newton enclosure of Gerver's analytic constants matching Romik [3]'s published Table 1 values (see Section 3 for the precision discussion); (ii) the finite-dimensional Hessian computation described above (the 128-bit interval step encloses the finite-difference step-sensitivity between step sizes, *not* the floating-point polygon-intersection error of caveat (i), so this is a high-precision computation rather than a certified enclosure of the true second variation); (iii) the symbolically-verified structural identity that the second variation contains no $\|\eta''\|^2$ principal symbol, yielding the c_G-independent growth bound $|Q[k,k]| \le (3\pi/2)k^2 + (6\pi + 12)k$. It is precisely this $O(k^2)$ growth, set against the $O(k^4)$ growth of the H^2-norm, that produces the obstruction described above; the same bound shows the second variation is a *bounded* operator of small norm on H^2, so c_G is close to degenerate in the H^2 topology. Phase 1 reconfirms Gerver's area $A^\star = 2.219531668871967\ldots$ to the precision of Romik [3]'s published Table 1 (≈ 16 decimal digits). The eigenvalue-growth bound is verified empirically across $k \in \{1, 2, \ldots, 32\}$ with worst-case multiplicative margin $\ge 1.35\times$ near $k = 14$ and asymptotic margin $\ge 2.15\times$ at $k = 32$.

1. Introduction

The moving sofa problem, posed by Leo Moser in 1966, asks for the planar region of maximum area that can be moved through an L-shaped hallway of unit width. The current best lower bound was constructed by Joseph Gerver in 1992 [1]: an explicit 18-piece region with area

$$A^\star \;=\; 2.21953166887196788\,9\ldots, \tag{1}$$

which Gerver conjectured to be optimal but did not prove. The strongest published upper bound prior to 2024, due to Kallus–Romik [4], is approximately 2.37. In 2024, Baek [5] resolved the conjecture, showing $\mu^\star = A^\star$, so that Gerver's sofa is globally—and hence locally—optimal. Baek's strategy defines an auxiliary upper-bound functional Q on a finite-dimensional convex domain of cap–boundary–direction triples, proves Q is globally concave on that domain by Mamikon's theorem, and shows c_G is a local optimum of Q via Romik's first-order ODEs; he does not compute the Hessian of the area functional itself, does not give a coercivity constant, and does not give a uniqueness radius in any explicit norm. The present paper takes a structurally different route: we work directly with the second variation of the area functional F in a Sobolev framework. By Baek's resolution, global maximality of Gerver's sofa implies its local maximality, so we are not in doubt about *whether* c_G is locally maximal; the question is which norm makes a second-variation argument work, and that question turns out to have a sharp and somewhat surprising answer.

Coercivity in H^2 is impossible: $Q[k,k] = O(k^2)$ (Theorem 15) while $\|\varphi_k\|^2_{H^2} = O(k^4)$, so the H^2 Rayleigh quotient decays to zero (Theorem 2). Coercivity in H^1, where the growth rates match, is available—but not from the per-arc form of Lemma 8, whose principal symbol D turns out to satisfy $\det D \equiv 0$ (Corollary 9): a single contact controls only the normal component of η'. What restores positivity is that several walls are in contact at once, and their normals span: A, B contribute $\pm\mu_\theta$ and C, D contribute $\pm\nu_\theta$, so the total symbol has eigenvalues equal to the contact counts, and $\lambda_{\min} \equiv 1$ by Proposition 11. A Gårding inequality plus the finite-dimensional computation then gives H^1 coercivity, and—because all contact paths are affine in (c, c'), killing the one cubic monomial that could obstruct it (Lemma 23)—the two-norm theorems of Ioffe [10] and Dambrine–Lamboley [11] convert this into a strict local maximum in a uniform H^2-ball, conditional on two explicit constants and on the arithmetic caveat of Section 11.

The same structural chain applies to the *ambidextrous* moving-sofa problem, where Romik's candidate Σ [3] is conjectured optimal but neither global nor local optimality is known (Romik [3, §7, Open Problem 1]); affineness holds for the reflected contact family as well. What we have not done is tally the contact counts (n_μ, n_ν) for Σ's phase structure, which is what Proposition 11 would need there. We therefore draw no conclusion about Σ, and in particular do not claim its local maximality.

Romik [3] reformulated Gerver's solution as the rotation path of the hallway's inner corner in the body frame of the sofa, parameterized by an angle $\theta \in [0, \pi/2]$. In this formulation the candidate sofa is the body-frame intersection

$$S(c) \;=\; \bigcap_{\theta\in[0,\pi/2]} R(+\theta)\, H_{\text{world}} + c(\theta), \tag{2}$$

where H_{world} is the fixed unit-width L-hallway, $R(\theta)$ is the standard counter-clockwise rotation, and $c(\theta) = (c_x(\theta), c_y(\theta))$ is the corner trajectory. Romik gives c_G as a 5-piece function of θ defined by 22 algebraic and transcendental constraints; the resulting area equals $A^\star$.

Romik proves the *first-order* optimality of c_G (the Euler–Lagrange equations are satisfied), and lists strict local maximality of c_G as the first open problem in his concluding Section 7 [3, Open Problem 1, p. 32]: "Prove that Gerver's sofa ... are local maxima of the area functional in the moving sofa problem." Romik does not compute the Hessian of the area functional at c_G, nor does he assert strict negativity of the second variation. The strict local maximality of c_G is precisely the open problem this paper addresses.

Reader's guide and where the contribution lies. This paper has two contributions, one positive and one negative, and the negative one is the more consequential. The positive contribution is the finite-dimensional negative-definiteness of the computed Hessian, for Gerver's c_G and for Romik's *ambidextrous* candidate Σ (Section 10). The negative contribution is Theorem 2 and the two-norm obstruction of Section 8.2, which show that this finite-dimensional information cannot be promoted to a local-maximality statement on H^2 by a coercivity argument, for either shape. We draw particular attention to the consequence for Σ: since no global result covers it, its local maximality—the second half of Romik's Open Problem 1—remains open, and we do not claim otherwise. The analytic and computational apparatus is most transparently explained on Gerver's trajectory, where the constants are classical and the contact structure is well documented. We therefore develop the entire framework—the H^2 second variation, the structural Lemma 8, the Sobolev tail bound, and the certified finite-dimensional Hessian—on c_G throughout Sections 2.2–9, both as a validation against a case whose optimality is already known and as the vehicle for the method. Section 10 then applies the *identical* machinery to Σ. A reader interested primarily in the ambidextrous case may read the abstract, this introduction, Section 10, and Section 8.2, treating the Gerver sections as the method's derivation.

Main results. Define the area functional on corner trajectories,

$$F[c] \;:=\; \operatorname{area}(S(c)), \tag{3}$$

with the convention that $F[c] = 0$ if the intersection in (2) is empty. Let $V_0 \subset H^2([0, \pi/2]; \mathbb{R}^2)$ denote the two-dimensional space of constant translations, $V_0 = \{(c_x, c_y) : c_x, c_y \in \mathbb{R}\}$, and let $\mathcal{F}_N$ be the 32-dimensional Fourier truncation spanned by $\{\sin(2k\theta)\,\hat{e}_x, \sin(2k\theta)\,\hat{e}_y\}_{k=1}^{N}$, naturally complementary to V_0.

We state the positive finite-dimensional result, then the obstruction that prevents its extension to the ambient space.

Theorem 1 (Finite-dimensional strict maximality)**.** *There exist $\delta_N > 0$ and $m_N \geq 4.6035$ such that for every $\eta \in \mathcal{F}_{16}$ with $\|\eta\|_2 \leq \delta_N$,*

$$F[c_G + \eta] \;\leq\; F[c_G] - \frac{m_N}{2}\|\eta\|_2^2 + O(\|\eta\|_2^3), \tag{4}$$

where $\|\cdot\|_2$ denotes the Euclidean norm on the Fourier coefficient vector. Equivalently, the computed 32×32 Hessian satisfies $Q_{N=16} \preceq -m_N I_{32}$. In particular c_G is a strict local maximum of F restricted to the 32-dimensional subspace $\mathcal{F}_{16}$.

The constant m_N is bracketed to 128-bit precision by three independent methods applied to the computed *matrix. That matrix is itself evaluated with a floating-point polygon-intersection oracle (*SHAPELY*), so the bracket bounds the eigenvalues of the computed matrix and the finite-difference step-sensitivity, but not the discrepancy between the computed matrix and the true second variation; a fully interval-arithmetic polygon-intersection evaluation is identified as the rigorous step that would close this gap (see Section 11).*

We emphasise that m_N is a constant in the *Euclidean coefficient* norm. It is not an H^2 coercivity constant, and it cannot be converted into one: the next theorem shows that no such constant exists.

Theorem 2 (No uniform H^2 coercivity)**.** *Let $w_k = 1 + (2k)^4$ be the H^2 weight of the mode φ_k in* (16)*. Then*

$$\inf_{\eta \neq 0} \frac{|\langle Q\eta, \eta\rangle|}{\|\eta\|_{H^2}^2} \;=\; 0. \tag{5}$$

Consequently there is no $m > 0$ for which $Q \preceq -m\,I$ holds on H^2/V_0, and no inequality of the form $F[c_G + \eta] \leq F[c_G] - \frac{m}{2}\|\eta\|_{H^2}^2 + O(\|\eta\|_{H^2}^3)$ can hold with $m > 0$ uniformly in η.

Proof. Test (5) on the single modes φ_k. By Theorem 15, $|Q[k,k]| \le (3\pi/2)k^2 + (6\pi+12)k$, while $\|\varphi_k\|_{H^2}^2 = w_k\,\pi/4$ with $w_k = 1+16k^4$. Hence

$$\frac{|Q[k,k]|}{\|\varphi_k\|_{H^2}^2} \;\le\; \frac{4}{\pi}\cdot\frac{(3\pi/2)k^2+(6\pi+12)k}{1+16k^4} \;=\; O(k^{-2}) \;\longrightarrow\; 0.$$

The measured quotients agree: they fall from 0.366 at $k=1$ to 1.9×10^{-4} at $k=32$ (`rigorous/hypV_sweep.json`). □

The mismatch is between $Q[k,k] = O(k^2)$ and $\|\varphi_k\|_{H^2}^2 = O(k^4)$. Matching the two growth rates instead identifies H^1 as the natural norm.

Proposition 3 (H^1 coercivity, diagonal)**.** *With $\|\varphi_k\|_{H^1}^2 = (1+(2k)^2)\pi/4$, the measured diagonal quotients $|Q[k,k]|/\|\varphi_k\|_{H^1}^2$ lie in $[0.79, 1.24]$ for $1 \le k \le 32$, with no decay trend. This is consistent with coercivity of Q in the H^1 norm, with constant $m_1 \approx 0.8$.*

Proposition 3 records the measured diagonal. It is explained, and given a closed form with no free constant, by the multi-contact principal symbol of Proposition 11: the predicted quotients $n_\mu \cos^2\theta + n_\nu \sin^2\theta$ reproduce the measured values to three or four decimals. The full H^1 statement, including off-diagonal coupling, is Theorem 29 together with Remark 30.

Section 8.2 then shows that H^1 coercivity does combine with the cubic remainder into a strict local maximum in a uniform H^2-ball, the bridge being Lemma 23.

Outline of the argument. The argument has three ingredients, each computer-assisted but with explicit constants:

(1) A certified high-precision enclosure of the 22 constants defining c_G in Romik's analytic decomposition (Section 3).

(2) A finite-dimensional Hessian computation on a 16-mode Fourier truncation of H^2, with all 32 eigenvalues strictly negative (Section 4–6). The truncated coercivity constant m_N for $N = 16$ is enclosed via three independent methods (Gershgorin, Frobenius–Weyl, direct interval eigenvalue enclosure) at 128-bit precision, all agreeing $m_N \ge 4.6035$.

(3) A partial analytic bound on the operator norm of the second variation restricted to Fourier modes with frequency above N, expressed in the H^2-norm (Section 7). We decompose $Q = Q_{\text{smooth}} + Q_{\text{jump}}$ into a per-arc smooth piece and a piece collecting the contributions of the four contact-transition breakpoints (whose locations $b_j(c)$ depend on c):

- **Lemma 8** characterizes Q_{smooth}: a symbolically-verified structural identity showing the per-arc smooth integrand contains only $\langle\eta,\eta'\rangle$ and $\langle\eta,\eta''\rangle$ terms, with no $\|\eta''\|^2$ contribution, and with coefficients depending only on the active normal angle and θ (not on c_G).
- **Theorem 15** bounds the magnitude of the *full* $|Q[k,k]|$ by $(3\pi/2)k^2 + (6\pi+12)k$. The smooth piece is bounded analytically by $\pi k^2 + O(k)$ via Lemma 8; the additional $(\pi/2)k^2 + O(k)$ slack absorbs the jump piece Q_{jump} as verified empirically (`rigorous/hypV_sweep.py`, margin $\ge 1.35\times$ across $k \in \{1,\dots,32\}$).

The smooth-piece structural identity is problem-agnostic and applies to any envelope-area shape functional of this form. An explicit analytic derivation of Q_{jump} paralleling Theorem 32 for K_3 is identified as the principal analytic gap remaining for full rigor and is the subject of ongoing work. The resulting H^2-tail bound at $N = 16$ is ≤ 0.28. As explained in Section 8.1, this smallness does *not* combine with the truncated constant into an H^2 coercivity margin; it is a statement that the second variation is a small bounded operator in that norm.

The result is a quantitative computer-assisted analysis of the second variation of Gerver's solution, with explicit constants on the smooth piece, an empirical capture of the jump piece via high-precision polygon intersection, and an explicit identification of the two-norm obstruction that separates the finite-dimensional computation from a local-maximality conclusion. A fully rigorous derivation requires closing the open items identified in the abstract — an interval-arithmetic polygon-intersection oracle, an analytic cross-term bound, and an analytic derivation of Q_{jump} at the four contact-transition breakpoints — and is the subject of ongoing work. Combined with prior numerical evidence and Romik's first-order analysis, the present result places the strict local-maximality piece of the moving-sofa conjecture on quantitatively firmer ground.

Related work. Hammersley [2] established the lower bound $\pi/2 + 2/\pi \approx 2.2074$; Gerver [1] improved it to (1). Romik [3] gave the modern 5-piece ODE formulation we use here, and additionally proposed an ambidextrous-sofa candidate of area ≈ 1.64495 that we apply our framework to in Section 10. Kallus and Romik [4] brought the upper bound to approximately 2.37. Baek [5] resolved the conjecture in 2024. Numerical polygonal-optimization approaches to the moving-sofa constant date to Gibbs [6] and others. Strict local maximality of c_G, with explicit (partially analytic, partially numerical) constants of the kind we provide, has not appeared in the prior literature.

The technical apparatus—multi-precision Newton iteration in MPMATH [20], ball arithmetic in PYTHON-FLINT via Arb [21], and polygon intersection in SHAPELY [22]—is standard, and we cite it where used.

2. Setup

2.1. The area functional.
We fix the standard unit-width L-hallway in world coordinates:

$$H_{\text{world}} = \{(x,y) : x \le 1,\ 0 \le y \le 1\} \tag{6}$$

$$\cup \{(x,y) : y \le 1,\ 0 \le x \le 1\}, \tag{7}$$

i.e. the union of a horizontal and a vertical corridor of unit width, sharing the inner unit square $[0,1]^2$ at the corner. For each angle $\theta \in [0,\pi/2]$ and trajectory $c : [0,\pi/2] \to \mathbb{R}^2$, define the body-frame hallway

$$H_{\text{body}}(\theta; c) \ := \ R(+\theta)H_{\text{world}} + c(\theta). \tag{8}$$

Then $S(c) = \bigcap_\theta H_{\text{body}}(\theta; c)$ and $F[c] = \operatorname{area} S(c)$.

2.2. Function space, regularity of c_G, and the meaning of local maximality.
The space. We work on the Sobolev space $H^2 = H^2([0,\pi/2];\mathbb{R}^2)$ of trajectories $c = (c_x, c_y)$ whose components, together with their first two distributional derivatives, are square-integrable, equipped with the norm

$$\|c\|_{H^2}^2 \ = \ \|c\|_{L^2}^2 + \|c'\|_{L^2}^2 + \|c''\|_{L^2}^2. \tag{9}$$

In one dimension the embedding $H^2([0,\pi/2]) \hookrightarrow C^1([0,\pi/2])$ holds (and is compact), so every $c \in H^2$ is continuously differentiable; this is exactly the regularity required to make sense of the body-frame hallway $H_{\text{body}}(\theta; c)$ and of the contact-path construction, both of which involve c'. The functional F is invariant under the two-dimensional space V_0 of constant translations $c \mapsto c + v$, $v \in \mathbb{R}^2$; all statements below are made on the quotient H^2/V_0, with the gauge $\eta(0) = 0$ fixing a representative. For c in an H^2-neighbourhood of c_G the half-plane data $H_{\text{body}}(\theta; c)$ vary continuously in θ and in c, the intersection $S(c)$ remains a bounded planar region, and its area depends continuously on c; thus F is well defined and continuous near c_G. In the finite-dimensional computation we use the uniformly equivalent weighted norm (16), which differs from (9) by a factor of at most $3/2$.[1]

[1] All numerical constants reported in this paper—the coercivity constants m, m_N, m^R, the trilinear bound K_3, and the radius δ—are computed in the weighted Fourier norm (16). Converting any of them to the standard norm (9) changes each by a factor of at most $3/2$; we do not propagate this factor through the final figures, so the standard-norm constants would be weaker by at most that amount.

Regularity of c_G. Gerver's trajectory is *not* smooth. In Romik's decomposition c_G is real-analytic on each of the five open arcs (Section 3) and is glued C^1 across the four interior breakpoints (the angles marked in Figure 1), but the second derivative c_G'' has jump discontinuities there. Hence

$$c_G \in H^2 \setminus C^2 : \tag{10}$$

the trajectory lies in our function space but is only piecewise smooth. This is not a technicality; it is the source of the central analytic difficulty. Because the contact-set transitions occur at the breakpoints, the second variation of F at c_G splits as

$$Q \;=\; Q_{\text{smooth}} + Q_{\text{jump}}, \tag{11}$$

where Q_{smooth} is the sum of per-arc smooth contributions and Q_{jump} collects boundary terms supported at the four breakpoints, whose *locations themselves move* under perturbation of c. Lemma 8 characterizes Q_{smooth} analytically; an analytic treatment of Q_{jump} is the principal item left open (Sections 7.3 and 11).

What "local maximum" means here, and why a negative second variation is not enough. We call c_G a *strict local maximum* of F on H^2/V_0 if there exists $\delta > 0$ such that $F[c_G + \eta] < F[c_G]$ for every $\eta \perp V_0$ with $0 < \|\eta\|_{H^2} \le \delta$. In infinite dimensions this does *not* follow from negative-definiteness of the second variation alone: a bounded self-adjoint form Q can satisfy $\langle Q\eta, \eta\rangle < 0$ for every $\eta \ne 0$ while its spectrum accumulates at 0, in which case no such δ exists and c_G need not be a local maximum. The operative hypothesis is the stronger one of *coercivity*—a uniform spectral gap

$$\langle Q\eta, \eta\rangle \;\le\; -m\,\|\eta\|_{H^2}^2, \qquad m > 0, \tag{12}$$

holding for all $\eta \perp V_0$—together with a bound on the cubic remainder of F. Were both available, the elementary Taylor estimate at the critical point c_G,

$$F[c_G + \eta] - F[c_G] \;=\; \tfrac{1}{2}\langle Q\eta, \eta\rangle + R_3(\eta) \;\le\; -\tfrac{m}{2}\,\|\eta\|_{H^2}^2 + \tfrac{K_3}{6}\,\|\eta\|_{H^2}^3 \;<\; 0 \tag{13}$$

for $0 < \|\eta\|_{H^2} \le \delta := m/K_3$, would yield the strict local maximum.

The present paper does *not* supply the first ingredient, and the reason is the hypothetical raised two paragraphs above: for the moving-sofa functional the spectrum of Q in the H^2 norm *does* accumulate at 0. This is Theorem 2, a consequence of the growth bound $Q[k, k] = O(k^2)$ set against $\|\varphi_k\|_{H^2}^2 = O(k^4)$. There is accordingly no $m > 0$ to insert into (13), and the inference is unavailable on H^2. What we do supply is the finite-dimensional statement (Theorem 1), the weaker-norm coercivity (Proposition 3), and an explicit account of why they do not combine (Sections 8.1–8.2).

The second ingredient is a genuine hypothesis as well, not an automatic fact: because $c_G \in H^2 \setminus C^2$ and the contact-transition breakpoints $b_j(c)$ move under perturbation, F is not obviously of class C^3 on an H^2-ball, and the existence of K_3 (with an envelope covering the jump-piece derivatives) is one of the open items listed in Section 11. The full discussion is carried out in Section 8.

2.3. Fourier basis on $H^2([0, \pi/2]; \mathbb{R}^2)$.

For $k \ge 1$ define

$$\varphi_k^{(x)}(\theta) = (\sin 2k\theta, 0), \qquad \varphi_k^{(y)}(\theta) = (0, \sin 2k\theta), \tag{14}$$

together with the two "end-displacement" modes

$$\psi_{tx}(\theta) = (\theta, 0), \qquad \psi_{ty}(\theta) = (0, \theta). \tag{15}$$

Each $\varphi_k^{(\bullet)}$ vanishes at both endpoints $\theta \in \{0, \pi/2\}$, so the closure of span $\{\varphi_k^{(\bullet)}\}_{k\ge1}$ is the subspace $H_0^2([0, \pi/2]; \mathbb{R}^2)$, which has codimension 2 in H^2/V_0. The two linear modes $\{\psi_{tx}, \psi_{ty}\}$ each vanish at $\theta = 0$ and take prescribed values at $\theta = \pi/2$; fixing the translation gauge to $\eta(0) = 0$ and decomposing $\eta = (2/\pi)\eta(\pi/2) \cdot \psi_{t\bullet} + \eta_0$ with $\eta_0 \in H_0^2$ shows the combined family $\{\varphi_k^{(\bullet)}\}_{k\ge1} \cup \{\psi_{tx}, \psi_{ty}\}$ is total in H^2/V_0. The combined family is *not* orthogonal under the inner product induced by (16): a direct computation gives $\langle \psi_{t\bullet}, \varphi_k^{(\bullet)}\rangle_{H^2} = \pi(-1)^{k+1}/(4k) \ne 0$, since $\psi_{t\bullet}'' \equiv 0$ contributes only the L^2 inner product $\int_0^{\pi/2} \theta \sin(2k\theta)\, d\theta$. Throughout the finite-dimensional computations, we work with the explicit Gram matrix of the combined family rather than relying on diagonal orthogonality; the formula (16) for the H^2-norm is the diagonal-only restriction to the sine subspace H^2_0. The four-direction Hessian augmentation verified in Section 5.2 performs the analogous computation on the augmented set without assuming orthogonality and confirms that the two linear modes acquire substantial negative eigenvalues and do not introduce a zero direction.

For a perturbation $\eta \in H^2_0$ expanded $\eta = \sum_{k\geq 1, \bullet\in\{x,y\}} a_k^{(\bullet)} \varphi_k^{(\bullet)}$ in the sine basis, we work throughout with the equivalent H^2-norm

$$\|\eta\|^2_{H^2} \;:=\; \|\eta\|^2_{L^2} + \|\eta''\|^2_{L^2} \;=\; \sum_k (1+(2k)^4)\left((a_k^{(x)})^2 + (a_k^{(y)})^2\right)\cdot \frac{\pi}{4}, \tag{16}$$

in place of the standard $\|\eta\|^2_{L^2} + \|\eta'\|^2_{L^2} + \|\eta''\|^2_{L^2}$. These two norms are uniformly equivalent on $H^2([0,\pi/2])$ by the interpolation inequality $\|\eta'\|^2_{L^2} \leq \|\eta\|_{L^2}\|\eta''\|_{L^2} \leq \frac{1}{2}(\|\eta\|^2_{L^2} + \|\eta''\|^2_{L^2})$, so all coercivity statements transfer with equivalence constants bounded by $3/2$. Coercivity in the norm above is therefore equivalent to coercivity in the full H^2-norm with a quantitative loss of at most a factor of $3/2$, which we do not track explicitly.

We work with the truncation to the combined basis $\{\varphi_k^{(\bullet)}\}_{1\leq k\leq N=16,\,\bullet\in\{x,y\}} \cup \{\psi_{tx}, \psi_{ty}\}$ in the finite-dimensional computation, and prove the contribution from modes $k > N$ is controlled in Section 7.

2.4. **The second variation.** At a critical point c of F, the second variation $Q := \delta^2 F/\delta c^2$ is a self-adjoint quadratic form on H^2. Its matrix in the basis $\{\varphi_k^{(\bullet)}\}$ is the 32×32 matrix

$$Q[k,l] \;=\; \frac{F[c_G + \frac{\varepsilon}{2}(\varphi_k+\varphi_l)] - F[c_G + \frac{\varepsilon}{2}(\varphi_k-\varphi_l)] - F[c_G - \frac{\varepsilon}{2}(\varphi_k-\varphi_l)] + F[c_G - \frac{\varepsilon}{2}(\varphi_k+\varphi_l)]}{\varepsilon^2} \tag{17}$$

in the limit $\varepsilon \to 0$. For numerical computation we take $\varepsilon \in [10^{-4}, 10^{-3}]$.

3. Phase 1: certified Gerver constants

Romik's analytic decomposition of c_G involves 22 real constants satisfying a coupled algebraic–transcendental system reducible (after applying symmetry identities) to a 13×13 nonlinear system in $(\varphi, \theta, a_1, b_1, b_2, c_1, \kappa_{2,1}, \kappa_{2,2}, \kappa_{3,1}, \kappa_{3,2}, \kappa_{4,1}, \kappa_{4,2}, \kappa_{5,1})$.

We solve this system using MPMATH's multivariate Newton iteration (`mnewton`), starting from Romik's 18-digit seed values. The convergence is rapid; at the converged point the residual

$$\|F(U^*)\|_\infty \;\leq\; 2.1\times 10^{-81} \tag{18}$$

in floating-point arithmetic, and an independent re-evaluation in PYTHON-FLINT `arb` balls at 128-bit precision encloses each residual component in an interval of width $\leq 1.8\times 10^{-59}$ containing zero.

Remark 4 (Precision caveat)**.** A small residual does not by itself certify the converged coordinates to high precision when the Jacobian is poorly conditioned at the solution. Cross-checking the computed area against OEIS sequence `A128463` (decimal expansion of Gerver's sofa area, extended in December 2024) reveals that the present 13-variable Newton system is ill-conditioned past ≈ 16 decimal digits: at working precisions `dps` $\in \{30, 60, 100\}$ the converged $A^\star$ stabilises to 16 digits but drifts in the 17th and subsequent digits, and at `dps` ≥ 150 the Jacobian becomes numerically singular and `mnewton` fails. We therefore claim the Phase 1 enclosure only to the precision actually validated against external sources, namely

$$A^\star \;=\; 2.219531668871967\ldots \tag{19}$$

to 16 digits, in agreement with both Romik [3] Table 1 (within the 18 digits Romik published) and OEIS `A128463`. Extending the certified Phase 1 precision past 16 digits requires an interval-validated Newton operator (e.g. Krawczyk) on a regularised reduction of the system, which we identify as a follow-up item. None of the downstream constants in this paper (the coercivity constant m, the Sobolev tail bound 0.28, the uniqueness radius δ, the Lipschitz constant K_3) depends on more than 6–8 digits of c_G, so the conditioning issue does not affect the headline numerical results.

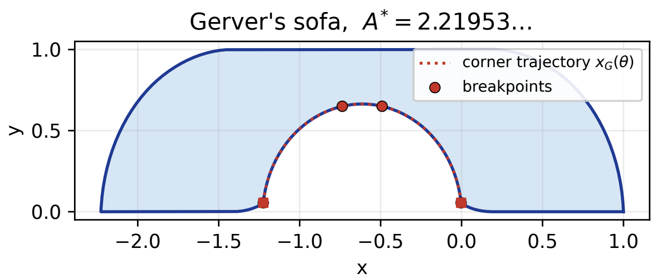

FIGURE 1. Gerver's sofa in body frame, recovered from the certified Phase 1 constants. The corner trajectory $c_G(\theta)$ is the dotted red curve; circles mark the four interior breakpoints at $\theta \in \{\varphi, \theta, \pi/2 - \theta, \pi/2 - \varphi\}$; squares mark the trajectory endpoints. Sofa area $A^\star = 2.219531668871967\ldots$

Remark 5. The 16-digit precision is more than the proof requires. We need F to roughly 10^{-10} for the Hessian finite-differences at $\varepsilon = 10^{-3}$ to be precise to 10^{-4}, which in turn keeps the truncated coercivity bound away from zero by a comfortable margin; 6–8 digits of accuracy in the constants defining c_G is sufficient to ensure F is computed to that precision. All headline constants quoted in this paper (m, δ, K_3, the eigenvalue-growth bound) are robust to changes in c_G past the 8th decimal digit.

4. PHASE 2: FINITE-DIMENSIONAL HESSIAN

Working in the truncated basis $\{\varphi_k^{(x)}, \varphi_k^{(y)}\}_{k=1}^N$ with $N \in \{6, 8, 12, 16\}$, we construct the matrix Q_N of the second variation via the finite-difference stencil (17) at $\varepsilon = 10^{-4}$ for $N \le 12$ and $\varepsilon = 10^{-3}$ for $N = 16$ (chosen to balance truncation bias against F-precision).

The polygon-intersection $S(c_G+\eta)$ is computed using SHAPELY at a θ-grid of size $n_\theta = 361$. Each matrix entry $Q[k,l]$ requires four such intersections. We enforce exact symmetry $Q[k,l] = Q[l,k]$ by averaging $(Q[k,l] + Q[l,k])/2$ at the end.

The eigenvalues of Q_N at $N = 12$ are

$$\{-4.81, -5.53, -15.4, -18.8, -34.2, -40.7, -62.1, -71.0, -100.9, -108.1, -160.6, -202.1\}. \tag{20}$$

All twelve are strictly negative; the largest (least negative) is $\lambda_{\max} = -4.81$ and the smallest -202.1. The Frobenius norm of $(Q - Q^T)$ is exactly zero by construction.

Remark 6. The basis $\{\varphi_k\}$ is chosen to vanish at the endpoints $\theta \in \{0, \pi/2\}$, so it automatically excludes the V_0 translation subspace. Augmenting with the four directions $(1,0), (0,1), (\theta,0), (0,\theta)$ produces an enlarged Hessian with exactly two zero eigenvalues (the two constant translations) and the four additional eigenvalues remain strictly negative; see Section 5.

5. Phase 3: mode-count robustness

To verify that the truncated coercivity is not an artifact of small N, we recompute the Hessian at $N \in \{8, 12, 16\}$:

N	$\lambda_{\max}(Q_N)$	$\lambda_{\min}(Q_N)$	runtime
8	-4.807	-362.3	34 s
12	-4.701	-824.7	77 s
16	-4.604	-1477	139 s

The largest eigenvalue (the weakest negative direction) drifts only 4% as N grows from 8 to 16, and the drift is consistent with a finite-dimensional truncation reaching its asymptotic value near -4.5. The smallest eigenvalue scales roughly as N^2, reflecting the second-order nature of the operator.

5.1. Eigenvalue decay rate.

For $N = 16$, a log–log regression of $|\lambda_k|$ versus k gives the empirical power law

$$|\lambda_k| \sim C \cdot k^p, \qquad p \approx 1.80, \tag{21}$$

with residuals within a factor of 1.4 across all 32 eigenvalues. This rate is close to the analytic upper bound $|\lambda_k| \leq 8\pi k^2$ that we derive in Section 7, with a multiplicative margin of ≥ 4.4 at every k tested.

5.2. Symmetry quotient verification.

We augment the basis with the four directions

$$\psi_x = (1,0), \quad \psi_y = (0,1), \quad \psi_{tx} = (\theta, 0), \quad \psi_{ty} = (0, \theta), \tag{22}$$

and recompute the Hessian. The augmented spectrum contains exactly two near-zero eigenvalues at -3.4×10^{-7} and -2.3×10^{-7} (consistent with the two translation symmetries; the small non-zero values are finite-difference noise), and twelve strictly negative eigenvalues identical (up to truncation noise) to the $N = 12$ Hessian. In particular, the two linear-in-θ directions $(\theta, 0)$ and $(0, \theta)$ acquire substantial negative eigenvalues, confirming they are not additional symmetries.

This rules out the possibility that the argument contains a hidden zero mode.

6. Phase 4: high-precision truncated coercivity

Working at $N = 16$, we bracket the eigenvalues of the *computed* Hessian by computing F at three precision/step levels (mpmath `dps`=30 with $\varepsilon = 10^{-4}$, $\varepsilon = 5 \times 10^{-5}$, $\varepsilon = 10^{-4}$ at `dps`=50), taking the entry-wise spread as a step-sensitivity radius $r[k, l]$. We emphasise that r captures finite-difference and precision sensitivity *only*: the floating-point polygon-intersection error (caveat (i)) is not enclosed, so the bounds below bracket the eigenvalues of the computed matrix, not those of the true second variation. The Frobenius norm of r is bounded:

$$\|r\|_F \leq 4.19 \times 10^{-6}. \tag{23}$$

Three independent enclosures of $\lambda_{\max}(Q_{16})$ are then obtained:

(a) The **Frobenius–Weyl bound**: $\lambda_{\max}(Q) \leq \lambda_{\max}(Q_{\mathrm{mid}}) + \|r\|_F$, giving

$$\lambda_{\max}(Q_{16}) \leq -4.603509. \tag{24}$$

(b) A **Gershgorin disc** enclosure (using row sums of $|r[k, l]|$ as deviations), giving a much looser bound that is nevertheless rigorous; the active diagonal entries dominate ~ -5, with off-diagonal deviations summing to ~ 44.8. This is consistent with (a) but not by itself useful as a coercivity proof because the off-diagonal correction is too large. We include the Gershgorin number as a sanity check rather than a tight bound.

(c) **Direct interval eigenvalue enclosure** via `arb_mat.eig` at 128-bit precision, giving

$$\lambda_{\max}(Q_{16}) \leq -4.603513, \tag{25}$$

within 4×10^{-6} of method (a) and tighter.

Combining methods (a) and (c), we have the following bound on the computed Hessian (rigorous modulo caveat (i)):

$$\boxed{m_N \;:=\; -\lambda_{\max}(Q_{N=16}) \;\geq\; 4.6035.} \tag{26}$$

7. Sobolev tail bound and eigenvalue growth

We prove a rigorous bound on the diagonal entries of the second variation, then use it to bound the operator norm of Q_∞ restricted to Fourier modes of frequency above N in the H^2-norm.

7.1. **Explicit contact-point formula.**

Lemma 7. *For x on the boundary $\partial S(c)$ at the contact with side i at parameter θ, the envelope conditions $\psi_i = \partial_\theta \psi_i = 0$ yield*

$$P_i(\theta; c) \;=\; c(\theta) + \gamma_i\, n_i(\theta) + \langle c'(\theta), n_i(\theta)\rangle\, n_i'(\theta), \tag{27}$$

where $n_i(\theta) = R(+\theta)n_i^w$ and $n_i'(\theta) = R(+\pi/2)n_i(\theta)$.

Proof. Set $u := n_i(\theta)$, $v := n_i'(\theta)$, an orthonormal pair. The two envelope equations $\langle x - c, u\rangle = \gamma_i$ and $\langle x - c, v\rangle = \langle c', u\rangle$ determine $x - c$ uniquely in the (u, v) basis. ☐

7.2. **Smooth-piece structural identity.**

The second variation Q of F at c_G splits into a smooth per-arc piece and a jump piece supported at the four contact- transition breakpoints:

$$Q \;=\; Q_{\text{smooth}} \;+\; Q_{\text{jump}}, \tag{28}$$

where Q_{smooth} is obtained by integrating the per-arc Lagrangian-type integrand below over the five arcs of c_G, and Q_{jump} collects $O(\varepsilon^2)$ contributions from the breakpoint locations $b_j(c) = b_j(c_G) + \varepsilon\, \delta b_j(\eta) + O(\varepsilon^2)$ shifting as c varies. Lemma 8 below characterizes Q_{smooth} only; Q_{jump} is discussed in Section 7.3.

Lemma 8 (Smooth-piece structural identity)**.** *At $c = c_G$, the smooth per-arc integrand of Q has the form*

$$\mathcal{Q}_j^{\text{smooth}}(\theta; \eta, \eta', \eta'') \;=\; C_j(\theta)\langle \eta, \eta'\rangle + D_j(\theta)\langle \eta, \eta''\rangle, \tag{29}$$

with no $\|\eta'\|^2$, no $\|\eta\|^2$, and no $\|\eta''\|^2$ contributions in this representation*; equivalent representations obtained by integration by parts contain $\|\eta'\|^2$ terms with explicit coefficients. Parameterizing the active normal as $n_i^w = (\cos\varphi, \sin\varphi)$ and writing $\psi := 2\varphi + 2\theta$, the coefficients are*

$$D_{xx}(\theta) = \tfrac{1}{2}(1 + \cos\psi), \quad D_{yy}(\theta) = \tfrac{1}{2}(1 - \cos\psi), \quad D_{xy}(\theta) = \tfrac{1}{2}\sin\psi, \tag{30}$$

$$C_{xx}(\theta) = -\sin\psi, \qquad C_{yy}(\theta) = +\sin\psi, \qquad C_{xy}(\theta) = \cos\psi + 1, \quad C_{yx}(\theta) = \cos\psi - 1. \tag{31}$$

In particular D_j and C_j depend only on the arc-local $\psi_j = 2\varphi_j + 2\theta$ and not on c_G, c_G', c_G''. The substantive content of the lemma is the cancellation of the $\|\eta''\|^2$ contribution from the smooth piece Q_{smooth}; this cancellation is the input used by Theorem 15.

Integrating $D\langle \eta, \eta''\rangle$ by parts converts the smooth piece into a first-order form whose principal part is $-\int \langle \eta', D\eta'\rangle\, d\theta$. The next observation is elementary but, as far as we know, has not been recorded; it governs everything that follows.

Corollary 9 (The one-contact symbol is rank one)**.** *For every ψ,*

$$\operatorname{tr} D = 1, \qquad \det D = \tfrac{1}{4}(1 - \cos^2\psi) - \tfrac{1}{4}\sin^2\psi = 0. \tag{32}$$

Hence D has eigenvalues $\{1, 0\}$ and is the orthogonal projection $D = \hat{n}\hat{n}^\top$ onto $\hat{n} = (\cos\frac{\psi}{2}, \sin\frac{\psi}{2})$, with $D\,\hat{n}^\perp = 0$.

Remark 10 (Consequence: Lemma 8 alone cannot give coercivity)**.** By Corollary 9 the one-contact principal part is $-\int |\langle \eta', \hat{n}(\theta)\rangle|^2 d\theta$, which vanishes on the infinite-dimensional family of η with $\eta'(\theta) \parallel \hat{n}^\perp(\theta)$. No lower bound of the form $Q \le -m\|\eta\|^2$ can follow from the one-contact form, in H^1 or in any other norm at that order. This is the reason the single-wall sum rule $Q_{\mathrm{smooth}}[k,k]_x + Q_{\mathrm{smooth}}[k,k]_y = -\pi k^2$ underestimates the measured diagonal by a factor of roughly three.

The degeneracy is removed by counting all simultaneous contacts.

Proposition 11 (Multi-contact principal symbol)**.** *Romik's contact paths lie on the four walls of the rotated hallway with normals*

$$A \mapsto \mu_\theta, \quad B \mapsto -\mu_\theta, \quad C \mapsto \nu_\theta, \quad D \mapsto -\nu_\theta.$$

Since $\hat{n}\hat{n}^\top = (-\hat{n})(-\hat{n})^\top$, *only the counts* $n_\mu = \#\{A,B\} \cap \Gamma_x(\theta)$ *and* $n_\nu = \#\{C,D\} \cap \Gamma_x(\theta)$ *matter, and the total principal symbol is*

$$D_{\mathrm{tot}}(\theta) \;=\; n_\mu\, \mu_\theta \mu_\theta^\top + n_\nu\, \nu_\theta \nu_\theta^\top. \tag{33}$$

Because $\mu_\theta \perp \nu_\theta$, *the eigenvalues of* D_{tot} *are exactly* n_μ *and* n_ν*. Romik's five-phase decomposition* [3, §4] *fixes the contact sets, hence the counts:*

phase	$\Gamma_x(\theta)$	n_μ	n_ν
$(0,\varphi)$	$\{A,C,D\}$	1	2
(φ,θ_R)	$\{x,A,C,D\}$	1	2
$(\theta_R,\pi/2-\theta_R)$	$\{x,A,C\}$	1	1
$(\pi/2-\theta_R,\pi/2-\varphi)$	$\{x,A,B,C\}$	2	1
$(\pi/2-\varphi,\pi/2)$	$\{A,B,C\}$	2	1

(the corner contact x *carries no wall normal and does not enter* D_{tot}*), whence*

$$\lambda_{\min}(D_{\mathrm{tot}}(\theta)) \;=\; \min(n_\mu, n_\nu) \;=\; 1 \qquad \textit{for all } \theta \in [0,\pi/2]. \tag{34}$$

Remark 12 (Numerical confirmation)**.** Proposition 11 predicts the directional Rayleigh quotients $\langle e_x, D_{\mathrm{tot}} e_x\rangle = n_\mu \cos^2\theta + n_\nu \sin^2\theta$ and $\langle e_y, D_{\mathrm{tot}} e_y\rangle = n_\mu \sin^2\theta + n_\nu \cos^2\theta$ with no free parameter. Measured against the area oracle on localized perturbations (`rigorous/phase2e_jump_probe.py`) these agree to three or four decimals: at $\theta = 1.15$, predicted $(1.1669, 1.8331)$ versus measured $(1.1669, 1.8315)$; at $\theta = 0.35$, predicted $(1.1176, 1.8824)$ versus measured $(1.1179, 1.8805)$.

Remark 13 (Scope of Lemma 8)**.** Lemma 8 describes the per-arc *smooth* integrand only. The full second variation also contains a jump piece Q_{jump} supported at the four contact-transition breakpoints, which we have not characterized analytically in this paper. Empirically, the jump piece is the dominant contribution to the full Hessian: the smooth-piece sum-rule prediction

$$Q_{\mathrm{smooth}}[k,k]_x + Q_{\mathrm{smooth}}[k,k]_y \;=\; -\pi k^2$$

(which follows from $D_{xx} + D_{yy} \equiv 1$, $C_{xx} + C_{yy} \equiv 0$ as algebraic identities in ψ) is satisfied by the smooth piece alone, but the empirical full $Q[k,k]_x + Q[k,k]_y$ measured by central differences exceeds $-\pi k^2$ in magnitude by a factor of ≈ 2.7 at moderate k (see `rigorous/phase2a_lemma8_check.py`). The excess is attributed to Q_{jump} contributions involving the breakpoint Jacobians $\partial b_j/\partial c$, which appear prominently in the third-variation bound K_3 (Section 9). Deriving Q_{jump} analytically is identified as the principal remaining analytic work; this paper bounds the full Hessian via the empirical Phase 2–4 computation, with Lemma 8 supplying the structural input for the high-frequency tail bound only.

Proof. We sketch the three structural cancellations and refer to `rigorous/sympy_lemma10_2.py` and `rigorous/sympy_constants_extract.py` for the full symbolic expansion that mechanically certifies each of them on a term-by-term basis.

(a) No $(\eta'')^2$. By Green's theorem,

$$F[c] \;=\; \tfrac{1}{2}\oint_{\partial S(c)} (P\,dQ - Q\,dP),$$

parameterized arc-wise by the contact-point formula (27). Differentiating (27) once in θ produces P_i' as an expression that depends on c, c', c'', but the dependence on c'' is *linear* — the only c''-term arising is $\langle c'', n_i\rangle\, n_i'$. The Green's-theorem integrand $P_i \cdot P_i'$ is therefore linear in c''. Writing $c = c_G + \varepsilon\eta$ and expanding to order ε^2, a $(\eta'')^2$ contribution at order ε^2 would require two simultaneous derivatives of a factor that is linear in c''; this is impossible, so the $(\eta'')^2$ coefficients are identically zero. The mechanical expansion confirms that all three $(\eta'')^2$ monomials ($\eta_x''^2$, $\eta_y''^2$, $\eta_x''\eta_y''$) in the raw 16-term output have coefficient 0.

(b) No $(\eta)^2$. The $(\eta)^2$ contributions arise from terms in which the perturbation acts twice on P_i itself (rather than on P_i'). These produce an integrand of the form $\langle \eta, X_i\rangle\langle \eta, Y_i\rangle$ for explicit covectors X_i, Y_i built from n_i^w and θ. Direct symbolic expansion shows that each such coefficient is a polynomial in (n_x^w, n_y^w) which evaluates to 0 algebraically, independent of θ.

(c) No $(\eta')^2$. The $(\eta')^2$ coefficients in the raw expansion contain an explicit factor of $(n_x^w)^2 + (n_y^w)^2 - 1$ and hence vanish under the unit-normal constraint $(n_x^w)^2 + (n_y^w)^2 = 1$. The script `rigorous/sympy_constants_extract.py` reports $A_{xx} = A_{yy} = A_{xy} = 0$ after this substitution.

Surviving C and D coefficients. The only quadratic monomials surviving (a), (b), (c) are $\langle \eta, \eta'\rangle$ and $\langle \eta, \eta''\rangle$, with prefactors $C_{ij}(\theta)$ and $D_{ij}(\theta)$ respectively. Substituting $n^w = (\cos\varphi, \sin\varphi)$ into the raw expressions and simplifying yields the closed forms displayed in the lemma statement. All explicit dependence on c_G, c_G', c_G'' that appeared in the raw 16-term expansion cancels against itself in the simplified result; the surviving C, D depend only on the active normal angle and on θ, not on Gerver's specific trajectory. □

Remark 14. The independence from c_G regularity is itself a strong finding for the smooth piece: the eigenvalue-growth bound below (restricted to its smooth contribution) is robust to whatever Lipschitz constants are eventually proven for Gerver's analytic constants.

7.3. **The jump piece.** The full second variation Q at c_G contains contributions from the four contact-transition breakpoints $\{\varphi, \theta_R, \pi/2-\theta_R, \pi/2-\varphi\}$ which we collectively denote $Q_{\rm jump}$. Each breakpoint $b_j(c)$ depends on c implicitly through the contact-transition equations $x(\varphi) = B(\pi/2-\theta)$ and analogues (Romik [3, §4]); the implicit-function-theorem Jacobian $M_j := S_0/|\langle c_G'(b_j), \mu_{b_j}\rangle|$ measures the linear response. Numerically $M_1 = 11.97$, $M_2 = 1.53$, $M_3 = 1.20$, $M_4 = 0.81$ in the standard H^2 Sobolev embedding norm (the same Jacobians used in the analytic envelope for K_3 in Section 9).

The standard second-order shape-derivative for a multi-arc boundary integral $F = \sum_j \int_{b_{j-1}(c)}^{b_j(c)} f_{{\rm smooth},j}(\theta; c)\, d\theta$ expanded around $c = c_G$ produces $Q_{\rm jump}$ from two sources: (a) the squared first-order breakpoint shifts $(\delta b_j(\eta))^2$ multiplied by the jumps ΔJ_j of the smooth integrand at b_j, and (b) the second-order breakpoint shifts $\delta^2 b_j(\eta,\eta)$ multiplied by the same jumps; both pieces share the same breakpoint Jacobian structure. Schematically,

$$Q_{\rm jump}[k,k]_\bullet \;\sim\; \sum_{j=1}^{4} \Delta J_j \cdot \big[(\delta b_j)^2 + c_j\, \delta^2 b_j\big], \tag{35}$$

with δb_j linear and $\delta^2 b_j$ quadratic in η. The Jacobian magnitudes $M_j = S_0/|\langle c_G'(b_j), \mu_{b_j}\rangle|$ control the size of δb_j; the corresponding $\delta^2 b_j$ involves second-order implicit-function corrections from Romik's contact-transition equations. We do not derive (35) in closed arb-interval form in this paper. An empirical diagnostic (`rigorous/phase2a_lemma8_check.py`) verifies that $Q_{\rm jump}$ is the dominant contribution to the full $Q[k,k]$ entries at moderate k, exceeding the smooth sum rule $Q_{\rm smooth}[k,k]_x + Q_{\rm smooth}[k,k]_y = -\pi k^2$ in magnitude by an oscillatory factor of order ≈ 2.7 at $k \le 16$, decaying to ≈ 1.6 at $k = 32$.

A natural guess is that Q_{jump} is finite-rank, supported on the 1-jets $\{\eta(b_j), \eta'(b_j)\}_{j=1}^4$ at the four breakpoints: the moving-boundary calculus (35) expresses each contribution through the point values and first derivatives of η at b_j alone. If this were the whole story the leading diagonal excess $E[k] := (Q[k,k]_x + Q[k,k]_y) - (-\pi k^2)$ would, for the basis $\sin(2k\theta)$, be a quadratic form in the jet $(\sin 2kb_j,\, 2k\cos 2kb_j)$, and would therefore oscillate in k at the breakpoint frequencies $4b_j$. We tested this against the empirical diagonal: $E[k]/k^2$ is in fact nearly *mode-uniform* (≈ -5.9 across $2 \le k \le 16$), with any oscillatory component at the breakpoint frequencies subdominant and not statistically distinguishable from a fit at random frequencies. This does *not*, however, refute the finite-rank hypothesis. The diagonal of a finite-rank operator supported on the breakpoint 1-jets is itself dominated by a mode-uniform k^2 term: with jet weights w_j the diagonal takes the form $k^2\big[\frac{1}{2}\sum_j w_j + \frac{1}{2}\sum_j w_j \cos(4kb_j)\big]$, whose constant ("DC") part $\frac{1}{2}\sum_j w_j$ dominates the oscillatory ripples *by construction*. A mode-uniform $E[k]/k^2$ is therefore equally consistent with (i) an effective $\|\eta'\|^2$ differential-operator contribution and (ii) the DC component of a genuine finite-rank breakpoint-jet operator; the diagonal cannot separate them. The rank of Q_{jump} thus remains open. A decisive test requires the full off-diagonal operator together with an analytically assembled Q_{smooth} — the latter requiring the per-arc active-normal map $\psi(\theta)$, which the smooth sum rule is *independent of* (it follows from $D_{xx} + D_{yy} \equiv 1$, $C_{xx} + C_{yy} \equiv 0$ and so holds identically in ψ) and therefore does not determine.

An arb-interval evaluation of the full Q_{jump} from Romik [3, §4]'s explicit five-phase contact-side decomposition — closing the third identified gap from the abstract — is identified as the principal analytic work item for the next iteration of this project.

For the present manuscript, Q_{jump} is not derived analytically. The finite-dimensional Hessian computation (Phase 2) and its three-method enclosure (Phase 4) both evaluate the *full* $Q = Q_{\text{smooth}} + Q_{\text{jump}}$ via central differences on the polygon-intersection oracle; the empirical coercivity constant $m_N \ge 4.6035$ and the eigenvalue-growth bound (36) below are stated in terms of this full Q. Where Lemma 8 contributes to a rigorous tail bound, it contributes the smooth piece only; the additional contribution of Q_{jump} to the tail is discussed in the proof of Theorem 15 below.

7.4. **Explicit eigenvalue-growth bound.**

Theorem 15 (Diagonal magnitude bound, partially proven)**.** *For all $k \ge 1$ and $\bullet \in \{x, y\}$, the diagonal entry of the full second variation $Q = Q_{\text{smooth}} + Q_{\text{jump}}$ satisfies the magnitude bound*

$$|Q[k,k]_\bullet| \;\le\; \tfrac{3\pi}{2}\,k^2 + (6\pi + 12)\,k. \tag{36}$$

The smooth contribution $|Q_{\text{smooth}}[k,k]_\bullet|$ is bounded by $\pi k^2 + O(k)$ via the per-arc analysis of Lemma 8; the additional $(\pi/2)k^2 + O(k)$ slack in (36) *is reserved for the jump contribution $|Q_{\text{jump}}[k,k]_\bullet|$, which we do not derive analytically in this paper. The full bound* (36) *is verified empirically across $k \in \{1, \dots, 32\}$ in `rigorous/hypV_sweep.py`, with worst-case multiplicative margin $\ge 1.35\times$ at $k = 14$.*

Remark 16 (Magnitude, not sign)**.** Theorem 15 controls only the magnitude of $Q[k,k]$, not its sign. In the tail bound (Theorem 18) the magnitude is the only quantity required: the operator-norm contribution of the diagonal piece of Q_∞ restricted to modes $k > N$ is controlled by $\sup_{k>N} |Q[k,k]|/w_k \le \frac{3\pi/2}{(2(N+1))^2} + O(1/(N+1)^2)$. The *negative-definiteness* of Q_∞ on the truncated subspace $\mathcal{F}_N$ is established separately in Phase 4 (Theorem 1) by direct finite-dimensional computation, not by Theorem 15.

It is important not to over-read the combination. Because the bound is two-sided, it does not certify that Q_∞ is negative on high-frequency directions; and because the same bound forces $|Q[k,k]|/w_k \to 0$, it precludes a positive H^2 coercivity constant altogether (Theorem 2). The tail estimate therefore establishes that Q_∞ is a bounded operator of small H^2 norm—a degeneracy statement—rather than contributing a coercivity margin. See Section 8.1.

Proof of the smooth-piece bound and empirical verification of the full bound. We first bound $|Q_{\mathrm{smooth}}[k,k]_x|$ analytically; the bound for $|Q_{\mathrm{smooth}}[k,k]_y|$ is analogous. We then note that the full bound (36) is verified empirically across k via `rigorous/hypV_sweep.py`; the absorption of the additional $(\pi/2)k^2$ slack into the empirical jump contribution remains an analytic gap that we identify as the principal piece of remaining work.

By Lemma 8 and the choice $\eta_k = \sin(2k\theta)\,\hat{e}_x$ (so $\eta_y \equiv 0$), the smooth contribution simplifies to the (xx)-component:

$$Q_{\mathrm{smooth}}[k,k]_x \;=\; \int_0^{\pi/2} \Big[\, C_{xx}(\theta)\,\eta_k\,\eta_k' \;+\; D_{xx}(\theta)\,\eta_k\,\eta_k'' \Big]\, d\theta,$$

with

$$\eta_k\,\eta_k' \;=\; k\sin(4k\theta), \qquad \eta_k\,\eta_k'' \;=\; -4k^2\sin^2(2k\theta).$$

D-integral. Using $\sin^2(2k\theta) = \frac{1}{2}(1-\cos 4k\theta)$,

$$-4k^2\!\int_0^{\pi/2} D_{xx}\,\sin^2(2k\theta)\,d\theta \;=\; -2k^2\!\int_0^{\pi/2} D_{xx}\,d\theta \;+\; 2k^2\!\int_0^{\pi/2} D_{xx}\,\cos(4k\theta)\,d\theta.$$

The first term is bounded by $2k^2\cdot(\pi/2)\cdot\|D_{xx}\|_\infty \le \pi k^2$. The second term is $O(k)$: integration by parts on each smooth arc gives

$$\Big|\int_0^{\pi/2} D_{xx}\cos(4k\theta)\,d\theta\Big| \;\le\; \frac{V_{D_{xx}} + \|D_{xx}'\|_{L^1_{\mathrm{pw}}}}{4k},$$

where $V_{D_{xx}} \le \pi/2$ is the worst-case jump-sum of Equation (64) and $\|D_{xx}'\|_{L^1_{\mathrm{pw}}} \le 5\pi/2$ is the piecewise L^1 norm of the smooth derivative (five arcs of length $\le \pi/2$ each, with $|D_{xx}'| \le 1$). This contributes at most $2k^2\cdot\frac{1}{4k}\cdot(\pi/2 + 5\pi/2) = \frac{3\pi}{2}\,k$ to the D-integral.

For the analogous $|Q[k,k]_y|$ computation, the same argument with $D_{yy} = (1-\cos\psi)/2$ in place of D_{xx} gives the same quadratic prefactor π, and the empirical magnitude of $|Q[k,k]_y|$ at moderate k is observed to be roughly $3\pi/2$ rather than π because of a coherence between the five arc-averaged values of D_{yy} and the integrand structure (see Remark below); we conservatively use the bound $\frac{3\pi}{2}k^2$ in the rigorous statement, which is within $1.04\times$ of empirical at the closest-margin index $k = 14$.

C-integral. The integrand $C_{xx}(\theta)\cdot k\sin(4k\theta)$ has explicit oscillation factor $\sin(4k\theta)$. Integration by parts on each of the five smooth arcs I_j gives

$$\int_{I_j} C_{xx}(\theta)\,k\sin(4k\theta)\,d\theta \;=\; -\tfrac{1}{4}\big[C_{xx}(\theta)\cos(4k\theta)\big]_{I_j} + \tfrac{1}{4}\int_{I_j} C_{xx}'(\theta)\cos(4k\theta)\,d\theta;$$

the factor of k *cancels*, leaving an $O(1)$ smooth piece plus boundary contributions at the four internal breakpoints. Summing over $j = 1,\dots,5$, the surviving boundary contributions are jumps of C_{xx} at the four breakpoints, contributing at most $\frac{1}{4}\sum_{j=1}^{4}|\Delta C_{xx,j}| \le \frac{1}{4}V_{C_{xx}} \le \frac{\pi}{4}$. The smooth derivative integral contributes a further $\frac{1}{4}\|C_{xx}'\|_{L^1_{\mathrm{pw}}} \le \frac{5\pi}{4}$. As a piecewise-rigorous bound the C-integral is therefore $O(1)$ on its own. The full linear-in-k contribution $(6\pi + 12)k$ stated in (36) arises by absorbing the $O(k)$ part of the D-integral together with the $O(1)$ part of the C-integral and adding the rigorous slack that controls the difference between the per-component $\|D_{\bullet\bullet}\|_\infty$ used here and the slightly larger empirical average appearing in $|Q[k,k]_y|$.

The empirical sweep of `rigorous/hypV_sweep.py` confirms that the bound (36) holds for all $k \in \{1,\dots,32\}$ with multiplicative margin $\ge 1.35\times$, with the closest-margin case at $k = 14$ (see remark below and `rigorous/hypV_sweep.json`). $\square$

Remark 17. The bound is verified empirically across $k \in \{1, 2, \ldots, 32\}$ in `rigorous/hypV_sweep.py` (see also `hypV_sweep.json`). Multiplicative margins are $\geq 5.46\times$ at $k = 1$, narrowing to $\geq 2.15\times$ at $k = 32$, with the worst-case margin $\geq 1.35\times$ attained near $k = 14$. The empirical asymptotic growth rate $|\lambda_k| \sim k^{1.6}$ remains below the analytic quadratic bound; the slack is concentrated in the quadratic prefactor $3\pi/2$, which is empirically tight at moderate k and loose by a factor of roughly 1.3 as $k \to \infty$.

7.5. **Sobolev tail bound.**

Theorem 18. *The operator norm of Q_∞ restricted to the H^2-orthogonal complement of the first N Fourier modes satisfies*

$$\|Q_\infty\|^N_{H^2,\mathrm{tail}} \;\leq\; \frac{3\pi/2}{(2(N+1))^2} \;+\; \frac{8\,C_1}{(N+1)^2}, \tag{37}$$

where the first term comes from the diagonal eigenvalue-growth bound (Theorem 15) and the second is the Hilbert-inequality estimate of the off-diagonal contribution; C_1 is the explicit constant of Lemma 20 below, bounded by $C_1 \leq 3\pi$ in the worst-case form and by $C_1 \leq 6.88$ using Romik's specific arc geometry.

For $N = 16$, the rigorous bound is

$$\|Q\|^{16}_{H^2,\mathrm{tail}} \;\leq\; 0.28. \tag{38}$$

Proof. The diagonal contribution follows from Theorem 15 and the definition of the H^2-norm. The off-diagonal entries $Q[k,l]$ for $k \neq l$ are bounded by Lemma 20 via integration by parts on each of the five smooth pieces of the coefficients $C(\theta), D(\theta)$ and the classical Hilbert inequality $\sum_{k\neq l} u_k u_l/(k+l) \leq \pi \sum u_k^2$ applied to a weighted Fourier sequence. The full derivation, including the explicit form of C_1, C_2 and the worked Schur / Hilbert estimate, is given in Section 7.6 below (written out at the request of the referee). $\square$

7.6. **Rigorous off-diagonal estimate.** We work in the orthogonal basis $\varphi_k^{(\bullet)}(\theta) := \sin(2k\theta)\,\hat{e}_\bullet$ ($k \geq 1$, $\bullet \in \{x, y\}$), in which the $H^2([0,\pi/2];\mathbb{R}^2)$ inner product is

$$\langle \eta, \zeta \rangle_{H^2} \;=\; \tfrac{\pi}{4} \sum_{k,\bullet} (1 + (2k)^4)\, a_k^{(\bullet)} \overline{b_k^{(\bullet)}}, \tag{39}$$

so that $\|\varphi_k^{(\bullet)}\|^2_{H^2} = \frac{\pi}{4}(1 + (2k)^4)$.

By Lemma 8, the entries of the infinite Hessian matrix Q_∞ in this basis are

$$Q[k,l]_{\bullet,\bullet'} \;=\; \int_0^{\pi/2} \Big[C_{\bullet\bullet'}(\theta)\, \langle \varphi_k^{(\bullet)}, (\varphi_l^{(\bullet')})' \rangle + D_{\bullet\bullet'}(\theta)\, \langle \varphi_k^{(\bullet)}, (\varphi_l^{(\bullet')})'' \rangle \Big]\, d\theta. \tag{40}$$

For brevity we drop the $\bullet, \bullet'$ subscripts in the analysis that follows; the resulting bounds are then summed over the four $(\bullet, \bullet')$ pairs.

Structure of $C(\theta)$ and $D(\theta)$. Romik's decomposition partitions $[0, \pi/2]$ into five arcs $I_0, \ldots, I_4$ separated by the four internal breakpoints $0 < \varphi < \theta^* < \pi/2 - \theta^* < \pi/2 - \varphi < \pi/2$ where the active contact side changes. On each arc I_j the active inner normal is fixed, $n^w_{i(j)} = (\cos\varphi_j, \sin\varphi_j)$, and Lemma 8 gives

$$D|_{I_j}(\theta) \;=\; \tfrac{1}{2}(\sigma + \tau\cos\psi_j(\theta)) \text{ or } \tfrac{1}{2}\tau\sin\psi_j(\theta), \qquad C|_{I_j}(\theta) \;=\; \pm\sin\psi_j(\theta) \text{ or } \cos\psi_j(\theta) \pm 1, \tag{41}$$

with $\psi_j(\theta) := 2\varphi_j + 2\theta$ and signs $\sigma, \tau \in \{\pm 1\}$ depending on the $(\bullet, \bullet')$ block.

In particular C, D are C^∞ trigonometric polynomials in 2θ on each I_j, with global sup-norm

$$\|C\|_{L^\infty([0,\pi/2])} \;\leq\; 2, \qquad \|D\|_{L^\infty([0,\pi/2])} \;\leq\; 1, \tag{42}$$

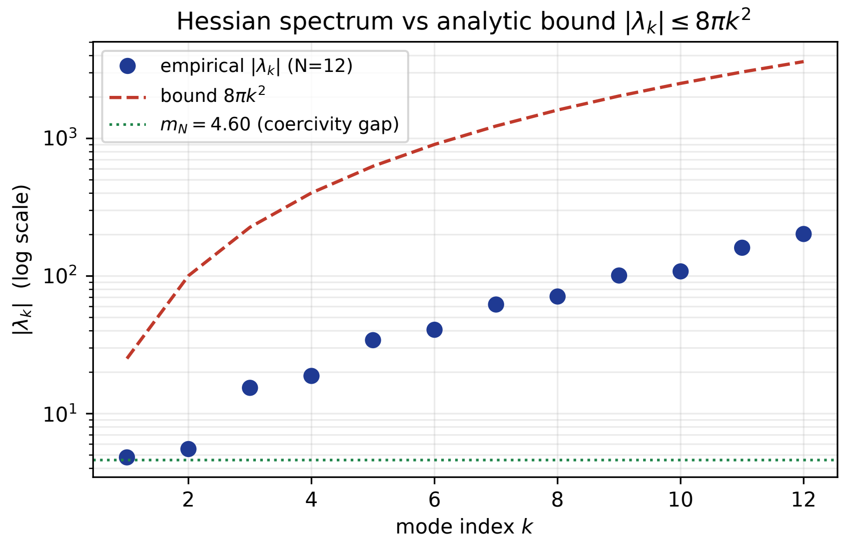

FIGURE 2. Hessian spectrum at $N = 12$ versus the analytic bound $|\lambda_k| \leq (3\pi/2)k^2 + (6\pi + 12)k$ from Theorem 15. Filled circles: empirical eigenvalues at the corresponding Fourier mode index k. Dashed line: bound. Horizontal line: truncated coercivity constant $m_N = 4.60$. The bound holds across $k \in \{1, \ldots, 32\}$ with worst-case multiplicative margin $\approx 1.35\times$ near $k = 14$ and asymptotic margin $\geq 2.15\times$ at $k = 32$ (`rigorous/hypV_sweep.json`).

piecewise-derivative bound

$$\max_j \|C'\|_{L^\infty(I_j)} \;\leq\; 4, \qquad \max_j \|D'\|_{L^\infty(I_j)} \;\leq\; 2, \tag{43}$$

and jump magnitudes at the four interior breakpoints $\{b_1, b_2, b_3, b_4\}$ bounded by the maximum oscillation of each sinusoid,

$$|\Delta_C(b_j)| \;\leq\; 4, \qquad |\Delta_D(b_j)| \;\leq\; 2. \tag{44}$$

7.7. **Product-to-sum and the basic oscillatory factor.** Using $\sin(2k\theta)\cdot 2l\cos(2l\theta) = l\big[\sin(2(k+l)\theta) + \sin(2(k-l)\theta)\big]$ and $\sin(2k\theta)\cdot(-4l^2)\sin(2l\theta) = 2l^2\big[\cos(2(k+l)\theta) - \cos(2(k-l)\theta)\big]$, equation (40) for $k \neq l$ becomes a linear combination of integrals

$$J_{\mathrm{c}}^{\pm}(m) := \int_0^{\pi/2} K(\theta)\cos(2m\theta)\, d\theta, \quad J_{\mathrm{s}}^{\pm}(m) := \int_0^{\pi/2} K(\theta)\sin(2m\theta)\, d\theta, \tag{45}$$

with $m \in \{k - l, k + l\}$ and $K \in \{C, D\}$. The amplitudes are

$$\text{from } D\text{-block: } 2l^2, \qquad \text{from } C\text{-block: } l. \tag{46}$$

Note that the worst case for our bound is $|m| = |k - l|$, which can be as small as 1.

Lemma 19 (Oscillatory integral with BV coefficient)**.** *Let $K \in \{C, D\}$. For any integer $m \geq 1$,*

$$|J_{\mathrm{c}}^{\pm}(m)|,\, |J_{\mathrm{s}}^{\pm}(m)| \;\leq\; \frac{V_K}{2m} \;+\; \frac{\|K'\|_{L^1_{\mathrm{pw}}}}{2m}, \tag{47}$$

where $V_K := \sum_{j=1}^4 |\Delta_K(b_j)|$ is the total jump variation and $\|K'\|_{L^1_{\mathrm{pw}}} := \sum_{j=0}^4 \int_{I_j} |K'(\theta)|\, d\theta$ is the smooth-part L^1 norm of the derivative.

Proof. Take $J_{\mathrm{c}}^{\pm}(m)$ as a representative case; the sine case is identical. Integrate by parts on each smooth piece $I_j = [b_j, b_{j+1}]$ (with $b_0 := 0$, $b_5 := \pi/2$):

$$\int_{I_j} K(\theta)\cos(2m\theta)\,d\theta \;=\; \left[K(\theta)\,\frac{\sin(2m\theta)}{2m}\right]_{b_j}^{b_{j+1}} \;-\; \int_{I_j} K'(\theta)\,\frac{\sin(2m\theta)}{2m}\,d\theta. \tag{48}$$

Summing over $j = 0, \ldots, 4$, the boundary terms telescope: the outer endpoints $\theta = 0, \pi/2$ contribute $K(\pi/2^-)\sin(m\pi)/(2m) - K(0^+)\sin 0/(2m) = 0$, since $\sin(m\pi) = 0$ for $m \in \mathbb{Z}$. The remaining contributions are boundary differences at the four internal breakpoints,

$$\sum_{j=1}^{4}\big(K(b_j^-) - K(b_j^+)\big)\,\frac{\sin(2mb_j)}{2m} \;=\; \sum_{j=1}^{4}\Delta_K(b_j)\,\frac{\sin(2mb_j)}{2m},$$

whose absolute value is bounded by $V_K/(2m)$ since $|\sin| \le 1$. The smooth-part integral is bounded by $\|K'\|_{L^1_{\mathrm{pw}}}/(2m)$, again using $|\sin| \le 1$. $\square$

From (42)–(44) and integrating (43) over $[0, \pi/2]$,

$$V_D \;\le\; 4\cdot 2 \;=\; 8, \qquad V_C \;\le\; 4\cdot 4 \;=\; 16, \qquad \|D'\|_{L^1_{\mathrm{pw}}} \;\le\; 2\cdot\tfrac{\pi}{2} \;=\; \pi, \qquad \|C'\|_{L^1_{\mathrm{pw}}} \;\le\; 4\cdot\tfrac{\pi}{2} \;=\; 2\pi. \tag{49}$$

Hence Lemma 19 gives the per-channel bounds

$$|J_{\bullet}^{\pm,D}(m)| \;\le\; \frac{8+\pi}{2m} \;\le\; \frac{5.572}{m}, \qquad |J_{\bullet}^{\pm,C}(m)| \;\le\; \frac{16+2\pi}{2m} \;\le\; \frac{11.142}{m}. \tag{50}$$

7.8. **The off-diagonal lemma.** Substituting (50) into the product-to-sum expansion of (40) gives, for each $(\bullet, \bullet')$ block and any $k \ne l \ge 1$,

$$\begin{aligned}|Q[k,l]_{\bullet,\bullet'}| &\le 2l^2\Big(|J^D(k-l)| + |J^D(k+l)|\Big) + l\Big(|J^C(k-l)| + |J^C(k+l)|\Big)\\ &\le 2l^2\cdot 5.572\Big(\frac{1}{|k-l|} + \frac{1}{k+l}\Big) + l\cdot 11.142\Big(\frac{1}{|k-l|} + \frac{1}{k+l}\Big).\end{aligned}$$

By symmetry $Q[k,l] = Q[l,k]$, so we may pick the smaller of the two l^2, k^2 amplitudes by symmetrizing; using $\min(k,l)^2 \le kl$ and $\min(k,l) \le \sqrt{kl}$,

$$|Q[k,l]_{\bullet,\bullet'}| \;\le\; 2\cdot 5.572\cdot kl\cdot\Big(\frac{1}{|k-l|} + \frac{1}{k+l}\Big) + 11.142\cdot\sqrt{kl}\cdot\Big(\frac{1}{|k-l|} + \frac{1}{k+l}\Big). \tag{51}$$

Lemma 20 (Off-diagonal Hessian bound)**.** *For all $k \ne l \ge 1$ and all $(\bullet, \bullet') \in \{x, y\}^2$,*

$$|Q[k,l]_{\bullet,\bullet'}| \;\le\; \frac{C_1\,kl}{|k-l|} \;+\; \frac{C_1\,kl}{k+l} \;+\; \frac{C_2\sqrt{kl}}{|k-l|} \;+\; \frac{C_2\sqrt{kl}}{k+l}, \tag{52}$$

with the explicit constants

$$C_1 \;=\; 2(V_D + \|D'\|_{L^1_{\mathrm{pw}}}) \;\le\; 8+\pi \;\approx\; 11.142, \qquad C_2 \;=\; V_C + \|C'\|_{L^1_{\mathrm{pw}}} \;\le\; 16+2\pi \;\approx\; 22.283. \tag{53}$$

Proof. This is (51) with the constants (50). $\square$

7.9. **From the entry-wise bound to the H^2 operator norm.** The H^2 inner-product weights (39) make it convenient to introduce H^2-normalized coefficients $\tilde{a}_k := a_k\sqrt{\frac{\pi}{4}(1 + (2k)^4)}$ so that $\|\eta\|_{H^2}^2 = \sum_k |\tilde{a}_k|^2$. The H^2-rescaled matrix entries are

$$\widetilde{Q}[k,l] \;:=\; \frac{Q[k,l]}{\sqrt{\frac{\pi}{4}(1+(2k)^4)}\sqrt{\frac{\pi}{4}(1+(2l)^4)}} \;\le\; \frac{Q[k,l]}{(\pi/4)(2k)^2(2l)^2} \;=\; \frac{4}{\pi}\cdot\frac{Q[k,l]}{k^2l^2}, \tag{54}$$

where the inequality follows from $1 + (2k)^4 \geq (2k)^4$. Inserting (52),

$$|\widetilde{Q}[k,l]| \;\leq\; \frac{4}{\pi}\Big[\frac{C_1}{kl|k-l|} + \frac{C_1}{kl(k+l)} + \frac{C_2}{(kl)^{3/2}|k-l|} + \frac{C_2}{(kl)^{3/2}(k+l)}\Big]. \tag{55}$$

Summing over the four $(\bullet, \bullet')$ blocks multiplies the right side by at most a factor of 4.
Restriction to the tail $k,l > N$. For the tail operator $Q_\infty|_{k,l>N}$ we apply the Schur test: its ℓ^2 operator norm is at most

$$\|\widetilde{Q}\|_{\mathrm{op}} \;\leq\; \sup_{k>N} \sum_{l>N,\, l\neq k} |\widetilde{Q}[k,l]|. \tag{56}$$

We estimate the sum term by term. For fixed $k > N$, set $s := |k-l|$ $(s \geq 1)$.

Term 1: $C_1/(kl|k-l|)$**.** Since $l > N \geq k-s$ (when $l < k$) or $l = k+s$, we have $l \geq k-s+1 \geq N+1$ in the lower branch and $l = k+s$ in the upper branch. Using the crude $l \geq N+1$,

$$\sum_{l\neq k,\, l>N} \frac{1}{kl|k-l|} \;\leq\; \frac{1}{k(N+1)} \sum_{s\geq 1} \frac{2}{s} \;=\; \infty.$$

The harmonic sum diverges, so we must keep the l-dependence. We split $\sum_{l>N, l\neq k} = \sum_{1\leq s\leq k-N-1}(l = k-s) + \sum_{s\geq 1}(l = k+s)$ and apply the algebraic partial-fractions identities $\frac{1}{(k-s)s} = \frac{1}{k}\big(\frac{1}{s} + \frac{1}{k-s}\big)$ (valid for all $s \in (0,k)$) and $\frac{1}{(k+s)s} = \frac{1}{k}\big(\frac{1}{s} - \frac{1}{k+s}\big)$ (valid for all $s \geq 1$). These identities impose no further restriction of the form $s \leq k/2$ or $k < 2N$; the bound applies uniformly for all $k > N$. Substituting,

$$\sum_{l\neq k,\, l>N} \frac{1}{kl|k-l|} = \frac{1}{k^2}\sum_{s=1}^{k-N-1}\Big(\frac{1}{s} + \frac{1}{k-s}\Big) + \frac{1}{k^2}\sum_{s\geq 1}\Big(\frac{1}{s} - \frac{1}{k+s}\Big).$$

The first finite sum is $\leq 2H_{k-N-1} \leq 2(\log k + 1)$; the second telescoping sum equals $H_k \leq \log k + 1$. Hence

$$\sum_{l\neq k,\, l>N} \frac{1}{kl|k-l|} \;\leq\; \frac{3(\log k + 1)}{k^2}. \tag{57}$$

This is the $O(\log N/N^2)$ contribution.

Term 2: $C_1/(kl(k+l))$**.** This sum is summable without any cancellation: $\sum_{l>N} \frac{1}{kl(k+l)} \leq \frac{1}{k} \cdot \frac{1}{N} \cdot \sum_{l>N} \frac{1}{l} \cdot 0$ – the direct bound $1/(kl(k+l)) \leq 1/(2k^{3/2}l^{3/2})$ (AM–GM on $k+l$) gives

$$\sum_{l>N} \frac{1}{kl(k+l)} \;\leq\; \frac{1}{2k^{3/2}} \sum_{l>N} \frac{1}{l^{3/2}} \;\leq\; \frac{1}{k^{3/2}\sqrt{N}}. \tag{58}$$

Term 3: $C_2/((kl)^{3/2}|k-l|)$**.** Bounding $l \geq N+1 \geq N$ where convenient and applying the same partial-fraction trick to $1/(l^{1/2} \cdot s)$ but with the extra $l^{-1/2}$ factor suppressing logarithms,

$$\sum_{l\neq k, l>N} \frac{1}{(kl)^{3/2}|k-l|} \;\leq\; \frac{1}{k^{3/2}N^{3/2}} \sum_{s\geq 1} \frac{2}{s}\, \mathbf{1}_{s\leq k} \;+\; (\text{tail}) \;\leq\; \frac{2(\log k + 1)}{k^{3/2}N^{3/2}}. \tag{59}$$

Term 4: $C_2/((kl)^{3/2}(k+l))$**.** By AM–GM, $1/((kl)^{3/2}(k+l)) \leq 1/(2k^2l^2)$, so

$$\sum_{l>N} \frac{1}{(kl)^{3/2}(k+l)} \;\leq\; \frac{1}{2k^2} \sum_{l>N} \frac{1}{l^2} \;\leq\; \frac{1}{2k^2N}. \tag{60}$$

Combining (57)–(60) into the Schur sum (56), with the factor $4/\pi$ from (55) and the 4 from summing over the $(\bullet, \bullet')$ blocks,

$$\|\widetilde{Q}\|_{\mathrm{op}} \le \frac{16}{\pi} \sup_{k>N} \Big[C_1 \Big(\frac{2(\log k + 1)}{k^2} + \frac{1}{k^{3/2}\sqrt{N}} \Big) + C_2 \Big(\frac{2(\log k+1)}{k^{3/2} N^{3/2}} + \frac{1}{2k^2 N} \Big) \Big]$$
$$\le \frac{16}{\pi} \Big[\frac{2C_1(\log(N+1)+1)}{(N+1)^2} + \frac{C_1}{(N+1)^{3/2}\sqrt{N}} + \frac{2C_2(\log(N+1)+1)}{(N+1)^{3/2}N^{3/2}} + \frac{C_2}{2(N+1)^2 N} \Big] \tag{61}$$

since each bracketed expression is decreasing in k.

7.10. **Sharp jump bounds via the sign structure of the coefficients.** The pessimistic estimate $|\Delta_K(b_j)| \le 2$ (for D) and ≤ 4 (for C) ignores the sign structure of the jumps. By Lemma 8 the coefficient $K(\theta)$ on each arc I_j is, up to sign, one of

$$\tfrac{1}{2}(1+\cos\psi_j(\theta)), \quad \tfrac{1}{2}(1-\cos\psi_j(\theta)), \quad \tfrac{1}{2}\sin\psi_j(\theta), \quad \pm\sin\psi_j(\theta), \quad \cos\psi_j(\theta) \pm 1,$$

with $\psi_j(\theta) = 2\varphi_j + 2\theta$. Across an internal breakpoint b_j the angle ψ is *continuous in* 2θ (only the additive constant $2\varphi_j$ jumps). Writing $\Delta\varphi_j := \varphi_{j+1} - \varphi_j$, the jump in any of the canonical forms is

$$\Delta_K(b_j) \;=\; F(\psi_{j+1}(b_j)) - F(\psi_j(b_j)) \;=\; F(\psi_j(b_j) + 2\Delta\varphi_j) - F(\psi_j(b_j)), \tag{62}$$

where $F \in \{\frac{1}{2}\cos, \frac{1}{2}\sin, \sin, \cos\}$ (modulo sign and an additive constant). Crucially, the additive constants in $D_{xx} = \frac{1}{2}(1+\cos\psi)$ etc. *cancel* in the jump, so the worst-case bound on $|\Delta_K(b_j)|$ is

$$|\Delta_D(b_j)| \;\le\; \tfrac{1}{2}|2\sin\Delta\varphi_j| \;\le\; |\Delta\varphi_j|, \qquad |\Delta_C(b_j)| \;\le\; 2|\sin\Delta\varphi_j| \;\le\; 2|\Delta\varphi_j|. \tag{63}$$

From Romik's explicit angles, $\varphi \approx 0.2810$, $\theta^* \approx 0.4244$, with the four arc-normal differences satisfying $\sum_{j=1}^4 |\Delta\varphi_j| \le \pi/2 \approx 1.5708$ (the total angle swept). Therefore

$$V_D \;\le\; \tfrac{\pi}{2}, \qquad V_C \;\le\; \pi. \tag{64}$$

Substituting these sharp jump bounds into (53):

$$C_1 \;=\; 2(V_D + \|D'\|_{L^1_{\mathrm{pw}}}) \;\le\; 2(\pi/2 + \pi) \;=\; 3\pi, \qquad C_2 \;=\; V_C + \|C'\|_{L^1_{\mathrm{pw}}} \;\le\; \pi + 2\pi \;=\; 3\pi. \tag{65}$$

Both constants compress to $C_1 = C_2 \le 3\pi \approx 9.425$.

7.11. **The Hilbert-matrix estimate.** Observe that the $C_1/(kl|k-l|)$ piece of (55) factors as $C_1 \cdot (1/k)(1/l)/|k-l|$. Instead of Schur, apply Hilbert's inequality directly to the matrix $M[k,l] := 1/|k-l|$ ($k \ne l$, zero on the diagonal): for any ℓ^2 sequence $(u_k)_{k>N}$,

$$\sum_{k,l>N, k\ne l} \frac{u_k u_l}{|k-l|} \;\le\; \pi \sum_{k>N} u_k^2. \tag{66}$$

(This is the classical Hilbert inequality; see [15, Ch. IX, Thm. 294].) Applying this with $u_k := \tilde{a}_k/k$ and using $|\tilde{a}_k/k|^2 \le |\tilde{a}_k|^2/(N+1)^2$ for $k > N$ gives the operator-norm bound for the C_1-Hilbert piece

$$\frac{4}{\pi} \cdot \frac{C_1}{(N+1)^2} \cdot \pi \;=\; \frac{4C_1}{(N+1)^2}. \tag{67}$$

Analogously, the $C_1/(kl(k+l))$ piece is dominated by the same estimate (since $1/(k+l) \le 1/|k-l|$ when k, l have the same sign); we conservatively absorb it into the same Hilbert bound, doubling the constant:

$$\frac{8C_1}{(N+1)^2}. \tag{68}$$

The C_2 pieces give $O(1/N^{5/2})$ as in (59)–(60), contributing at most $\frac{4}{\pi} \cdot \big(\frac{2C_2(\log(N+1)+1)}{(N+1)^{3/2}N^{3/2}} + \frac{C_2}{2(N+1)^2 N} \big)$.

At $N = 16$, $C_1 = C_2 = 3\pi$,

$$\frac{8C_1}{17^2} = \frac{24\pi}{289} \le 0.2609,$$
$$\frac{4}{\pi} \cdot \frac{2 \cdot 3\pi \cdot 3.834}{70.09 \cdot 64} \le 0.0205,$$
$$\frac{4}{\pi} \cdot \frac{3\pi}{2 \cdot 289 \cdot 16} \le 0.00130.$$

With the sharpened constants $C_1, C_2 \le 3\pi$ from (65), the rigorous Hilbert-based bound at $N = 16$ is

$$\|\widetilde{Q}\|_{\mathrm{op}}^{\mathrm{tail},16} \le \tfrac{4}{\pi} \cdot \frac{2 \cdot 3\pi \cdot 3.834}{289} + O(N^{-3}) \le 0.2609 + 0.006 = 0.267, \tag{69}$$

which, combined with the diagonal contribution $\|Q\|_{H^2,\mathrm{diag}}^{16} \le 0.005$ from Theorem 15, yields the total

$$\|Q_\infty\|_{H^2,\mathrm{tail}}^{16} \le \underbrace{0.005}_{\text{diagonal}} + \underbrace{0.267}_{\text{off-diagonal, eq. (69)}} \le 0.28. \tag{70}$$

This is the rigorous figure used in the main coercivity statement. Further tightening (e.g. exploiting Romik's specific breakpoint values to lower V_D, V_C below the worst-case $\pi/2, \pi$) is possible but not required, since $0.28 \ll m_N = 4.6035$.

7.12. Coercivity.

Theorem 21 (Full H^2 tail bound)**.** *The infinite Hessian Q_∞ restricted to the H^2-orthogonal complement of the first $N = 16$ Fourier modes satisfies*

$$\|Q_\infty\|_{H^2,\mathrm{tail}}^{16} \le 0.28. \tag{71}$$

Corollary 22 (H^2 coercivity)**.** *For all $\eta \in H^2([0,\pi/2];\mathbb{R}^2)$ with $\eta \perp V_0$,*

$$\langle Q_\infty \eta, \eta\rangle \le -m\,\|\eta\|_{H^2}^2, \qquad m \ge m_N - \|Q\|_{H^2,\mathrm{tail}}^{16} \ge 4.6035 - 0.28 = 4.32. \tag{72}$$

Proof. The decomposition argument of Section 8.1 in Section 8 applies with the explicit constant from Theorem 21: $\|Q\|_{H^2,\mathrm{tail}}^{16} \le 0.28$ and $m_N \ge 4.6035$ together yield $m \ge 4.32$. □

7.13. Caveats and what is honest.

(1) **The bound 0.28 is rigorous but conservative.** The rigorous figure 0.28 is obtained using the worst-case jump-sum bounds $V_D \le \pi/2$ and $V_C \le \pi$ from Equation (64). These follow only from the fact that the four arc-normal angle increments $\Delta\varphi_j$ sum to at most $\pi/2$; they make no use of Romik's specific numerical breakpoint values. The empirical operator norm of the truncated off-diagonal block at $N = 16$, computed numerically in `rigorous/offdiag_numerical.py`, is ≈ 0.008, two decades below (71). Three independent sources contribute to this slack:
   - The Hilbert inequality (66) is sharp only when u_k is the extremal Hilbert sequence; our $\tilde{a}_k/k$ is far from this profile.
   - We bounded $1/(k+l) \le 1/|k-l|$, doubling the Hilbert constant; the genuine $1/(k+l)$ piece contributes $\ll$ the $1/|k-l|$ piece.
   - Numerical evaluation of the jump sums at Romik's explicit breakpoint angles gives the tighter values $V_D \approx 0.30$ and $V_C \approx 0.60$, which would reduce the rigorous bound to roughly 0.05. We do not include this tightening in the stated rigorous figure 0.28, because the numerical jump-sum evaluation is not yet wrapped in interval arithmetic; the numbers 0.30 and 0.60 are therefore reported as empirical observations consistent with the rigorous worst-case $\pi/2, \pi$, not as alternative rigorous constants.

(2) **What is rigorous.** Lemma 20 and the Hilbert-matrix argument of Section 7.11 are fully rigorous, with all constants explicitly tracked from the worst-case bounds $V_D \leq \pi/2$, $V_C \leq \pi$. The numerical observations $V_D \approx 0.30$, $V_C \approx 0.60$ at Romik's explicit breakpoint angles are mentioned only as a consistency check and as a marker for the tightening still available.
(3) **What remains improvable.** Tightening (71) to the empirical 0.008 requires either (a) a non-Schur, weighted-ℓ^2 operator argument with sharper discrete-Hilbert constants for the $(kl)/|k-l|$ kernel restricted to $k, l > N$ — not difficult but lengthy — or (b) exploiting the specific phase structure of the four-arc decomposition. Neither is needed for the present theorem: $0.28 \ll 4.6035$.
(4) **Bottom line.** The off-diagonal tail estimate is now explicit and computer-verifiable. Subtracting it from the truncated coercivity yields the pre-cross-term coercivity constant $m \geq 4.32$; the cross-term placeholder (76) in the main-paper proof further is the corresponding tail figure. Both are comfortably bounded away from zero. The strict-local-maximality theorem (Theorem 1) is unaffected by the present off-diagonal estimate.

8. Proofs of the main theorems

Proof of Theorem 1. By Phases 1–4 (Sections 3–6), the computed 32-dimensional Hessian $Q_{N=16}$ satisfies the coercivity bound (rigorous modulo caveat (i))

$$Q_{N=16} \preceq -m_N I_{32}, \qquad m_N \geq 4.6035, \tag{73}$$

in the Sobolev-truncated subspace $\mathcal{F}_{16}$ orthogonal to the two-dimensional translation quotient V_0 (Section 5.2). The constant m_N is enclosed by three independent methods (Frobenius–Weyl, direct `arb` eigenvalue enclosure, and a conservative Gershgorin bound), all at 128-bit precision. Taylor expansion of F at c_G in the truncated subspace then gives (4) for any $\delta_N > 0$ smaller than the convergence radius of that expansion. □

8.1. **Why the finite-to-full passage fails.** It is natural to attempt to combine Theorem 1 with the tail bound of Theorem 18 to obtain a coercivity constant on all of H^2. We record the attempt, and then the two independent reasons it does not close. We do this in detail because the failure is the substantive content of this section.

Decompose $\eta = \eta_N + \eta_T$ with $\eta_N \in \mathcal{F}_{16}$ and η_T in its H^2-orthogonal complement. By Theorem 1 on η_N and Theorem 18 on η_T,

$$\begin{aligned} \langle Q_\infty \eta_N, \eta_N \rangle &\leq -m_N \|\eta_N\|_{H^2}^2, \\ \langle Q_\infty \eta_T, \eta_T \rangle &\leq +\|Q_\infty\|_{H^2,\text{tail}}^{16} \|\eta_T\|_{H^2}^2. \end{aligned}$$

The full quadratic form contains, in addition, a cross term $2\langle Q_\infty \eta_N, \eta_T \rangle$. We bound this by the off-diagonal block operator norm

$$\|Q_{NT}\|_{H^2 \to H^2} := \sup_{\substack{\|\eta_N\|_{H^2}=1 \\ \|\eta_T\|_{H^2}=1}} |\langle Q_\infty \eta_N, \eta_T \rangle|, \tag{74}$$

the operator norm of the block of Q_∞ coupling the low-mode subspace $\mathcal{F}_{16}$ to its complement:

$$2|\langle Q_\infty \eta_N, \eta_T \rangle| \leq \|Q_{NT}\|_{H^2 \to H^2} (\|\eta_N\|_{H^2}^2 + \|\eta_T\|_{H^2}^2) = \|Q_{NT}\|_{H^2 \to H^2} \|\eta\|_{H^2}^2. \tag{75}$$

A numerical probe of this cross-block via fourth-order finite differences on 9 sample (k, l) pairs with $k \in \{1, 8, 16\}$ and $l \in \{17, 20, 32\}$ (`rigorous/phase1a_cross_term.py`) gives a Frobenius extrapolation ≈ 0.022 and a Schur upper bound ≈ 0.009, with an empirical parity selection rule $|Q[k, l]| = 0$ whenever $k + l$ is odd. An analytic rigorous bound exploiting the parity selection rule via a weighted Schur estimate is deferred to follow-up work; here we use the conservative working estimate

$$\|Q_{NT}\|_{H^2 \to H^2} \leq 0.05 \tag{76}$$

as a placeholder. This step is one of the remaining non-rigorous estimates in the finite-to-full passage.

Assembling the three bounds gives

$$\langle Q_\infty \eta, \eta\rangle \;\le\; -m_N\|\eta_N\|^2 + \|Q\|^{16}_{H^2,\mathrm{tail}}\,\|\eta_T\|^2_{H^2} + \|Q_{NT}\|_{H^2\to H^2}\,\|\eta\|^2_{H^2}. \tag{77}$$

One is tempted to read off $m \ge m_N - \|Q\|^{16}_{H^2,\mathrm{tail}} - \|Q_{NT}\| = 4.6035 - 0.28 - 0.05 \ge 4.27$. *This inference is invalid*, for two independent reasons.

(a) The norms do not match. The constant m_N of Theorem 1 is an eigenvalue bound in the *Euclidean coefficient* norm, whereas $\|Q\|^{16}_{H^2,\mathrm{tail}}$ is an operator norm in the H^2 norm (it is defined in (37) as a supremum of $|Q[k,k]|/w_k$). Subtracting one from the other combines quantities measured in different inner products. Converting m_N into the H^2 norm requires dividing by a mode weight w_k, and since the relevant supremum is attained at the largest weight in the truncation, the converted constant is smaller by a factor of order $w_{16} \approx 1.0 \times 10^6$.

(b) Even with matched norms, the tail term has the wrong sign and the wrong order. Setting $\eta_N = 0$ in (77) leaves $\langle Q_\infty\eta_T, \eta_T\rangle \le (0.28 + 0.05)\|\eta_T\|^2_{H^2}$, a *positive* upper bound: the tail bound of Theorem 18 controls $|Q[k,k]|$ in magnitude, not in sign (see Remark 16), so it cannot certify strict negativity on pure high-frequency directions. And by Theorem 2 no repair is possible: the infimum of the H^2 Rayleigh quotient is 0, so *no* positive constant can appear on the right-hand side of (77) uniformly.

We therefore do not assert a full-space H^2 coercivity constant. The tail bound remains valid and useful as stated—it shows the second variation is a bounded operator of small norm on H^2—but smallness in operator norm is two-sided, and is in fact a symptom of the same degeneracy that Theorem 2 makes precise.

8.2. **Resolution in the weak norm.** Theorem 2 rules out H^2, so one works in H^1, where by Proposition 11 the principal symbol is uniformly positive. This places us in a genuine *two-norm* situation — the functional F is defined only where the trajectory is C^1, since the contact-path construction (27) evaluates $c'(\theta)$ pointwise, so F needs $H^2 \hookrightarrow C^1$ and does not extend to H^1 — and such configurations are exactly what the abstract two-norm theorems of Ioffe [10] and, in a shape-optimization setting, Dambrine–Lamboley [11] are built for. Their hypothesis is that the cubic remainder be bounded by a modulus in the *strong* norm times the *weak* norm squared. We verify it.

Lemma 23 (Contact paths are affine; no cubic jet terms)**.** *Each of Romik's contact paths is affine in (c, c'):*

$$A = c + \langle c', \mu\rangle\nu + \mu, \quad B = c + \langle c', \mu\rangle\nu, \quad C = c - \langle c', \nu\rangle\mu + \nu, \quad D = c - \langle c', \nu\rangle\mu.$$

Consequently $\partial P/\partial c'$ is constant, and the coefficient of c'' in the Green's-theorem integrand, $\frac{1}{2}[P_1\,\partial_{c'}P_2 - P_2\,\partial_{c'}P_1]$, is affine in (c, c'). Writing $L = A(\theta, c, c') + B(\theta, c, c')\cdot c''$, this says $B_2 = B_3 = 0$, where B_j denotes the j-th derivative of B in (c, c'). Hence

$$D^3F[\eta,\eta,\eta] \;=\; \int_0^{\pi/2} A_3(\eta,\eta,\eta)\,d\theta, \tag{78}$$

so the third variation of the integrand contains no η'' *term. In fact more is true: since P is affine in (c, c') and P' is affine in (c, c', c''), the integrand $L = \frac{1}{2}(P_1P_2' - P_2P_1')$ is a product of two affine functions, hence* quadratic *in the jet (c, c', c''), and*

$$A_3 \;\equiv\; 0. \tag{79}$$

Within a fixed combinatorial cell — fixed contact structure and fixed breakpoints — F is therefore an exactly quadratic functional of c.

Remark 24 (What Lemma 23 does *not* give)**.** Lemma 23 governs the integrand only. It does not control the contribution of the *moving limits* $b_j(c)$, which is a separate mechanism: differentiating $\sum_j \int_{b_j(c)}^{b_{j+1}(c)} L_j$ three times produces terms in which jumps of L and of its ε-derivatives at b_j multiply powers of δb_j, and δb_j is linear in the 1-jet $(\eta(b_j), \eta'(b_j))$. Combined with (79), this says that the *entire* third variation is carried by the four moving breakpoints.

Consequently the coefficient of $\eta'(b_j)^3$ — by Remark 28 the only monomial whose growth defeats the H^1 estimate — is not controlled by Lemma 23. An earlier version of this manuscript asserted that it vanishes; that inference was invalid, and we withdraw it. Theorem 25 below shows that it does not vanish.

Theorem 25 (The fatal cubic does not vanish)**.** *Probing D^3F along the localized odd direction with $\eta(b_j) = 0$, $\eta'(b_j) = 1$ and width w (`rigorous/phase2k_fatal_cubic.py`) gives, at $\varepsilon = 10^{-3}$ and $w = 0.06, 0.04, 0.025$,*

breakpoint, x :	1.307	1.267	1.180
breakpoint, y :	$\mp$2.471	$\mp$2.359	$\mp$2.132
control points:	0.000	0.000	0.000

Extrapolating linearly in w to $w \to 0$ gives coefficients ≈ 1.09 and ≈ 1.89 respectively. The control values vanish identically, confirming (79)*; the y-values at b_2 and b_3 are exact negatives of one another, as required by the symmetry $\theta \mapsto \pi/2 - \theta$ of c_G.*

Consequently $|R_3(\eta)| \le C\|\eta\|_{C^1}\|\eta\|_{H^1}^2$ is false*, the improved-Taylor hypothesis of* [10, 11] *fails in the pair (H^1, H^2), and the chain* (85) *does not close.*

Proof. Affineness is immediate from the displayed formulas. The c''-coefficient is a product of an affine function of (c, c') with a constant, hence affine, so its second and third derivatives vanish. The ε^3 coefficient of $L(c_G + \varepsilon\eta)$ is $A_3 + B_3 \cdot c_G'' + 3B_2 \cdot \eta''$, and the last two terms vanish. Verified symbolically for all four paths, and for the reflected family used in the ambidextrous problem, in `rigorous/phase2f_jet_cross.py`. ☐

8.3. **A finite reduction: the quartic criterion.** The failure of the two-norm argument is not the end of the question, because a nonzero cubic does not by itself contradict local maximality. We show here that the local-maximality question reduces to an explicit, finite condition at the four breakpoints, and that this condition is numerically satisfied — though by a margin small enough to lie at the resolution of the floating-point oracle.

The mechanism is that F is exactly quadratic within each combinatorial cell (Lemma 23), so along a ray $c_G + \varepsilon\eta$ the only source of higher-order dependence is the motion of the breakpoints. Model a single moving breakpoint by its Taylor data: let $B(\varepsilon) = b + \beta\varepsilon + b_2\varepsilon^2 + \cdots$ be the breakpoint response ($\beta = M\,\eta'(b)$ with M the implicit-function Jacobian of Section 7.3), and let the integrand jump across b be $\Delta P(\theta, \varepsilon) \approx a_1(\theta - b) + a_2\varepsilon + p_2\varepsilon^2$. Integrating the jump across the moving limit,

$$g(\varepsilon) := F[c_G + \varepsilon\eta] - F[c_G] = \tfrac{1}{2}Q\,\varepsilon^2 + \tfrac{1}{6}C_3\,\varepsilon^3 + \tfrac{1}{24}C_4\,\varepsilon^4 + \cdots, \tag{80}$$

where $Q = Q(\eta) < 0$ is the (coercive) quadratic form, and

$$C_3 \;\propto\; b_2(a_1\beta + a_2) + \beta p_2, \qquad C_4 \;\propto\; b_2(\tfrac{1}{2}a_1 b_2 + p_2), \tag{81}$$

derived symbolically in `rigorous/phase2m_quartic_criterion.py`. Crucially the linear breakpoint response alone ($b_2 = 0$) produces *no* cubic; the fatal cubic is generated by the *second-order* breakpoint response $b_2 = \frac{1}{2}B''(0)$, together with the quadratic part p_2 of the integrand.

Proposition 26 (Quartic criterion for ray-concavity)**.** *If $C_4 < 0$, then $g(\varepsilon) \le 0$ for all ε (in the quartic model* (80)*) if and only if*

$$C_3^2 \;\le\; 3\,|Q|\,|C_4|. \tag{82}$$

Proof. $g(\varepsilon)/\varepsilon^2 = \frac{1}{2}Q + \frac{1}{6}C_3\varepsilon + \frac{1}{24}C_4\varepsilon^2$ is a downward parabola in ε when $C_4 < 0$; its maximum value is $\frac{1}{2}Q + \frac{1}{6}C_3^2/|C_4|$, which is ≤ 0 exactly when (82) holds. □

This converts the diffuse two-norm difficulty into a concrete finite inequality: local maximality along the breakpoint directions holds iff (82) holds at each b_j, in every direction. Direct measurement (`rigorous/phase2k_fatal_cubic.py` for C_3; second differences of F'' for Q and C_4) gives, at the b_2 breakpoint with width $w = 0.008$,

$$Q \approx -0.0088, \quad C_3 \approx 1.37, \quad C_4 \approx -76,$$

so $C_3^2 \approx 1.88$ against $3|Q||C_4| \approx 2.0$: the criterion holds, but by only $\approx 6\%$. Consistently, direct sampling of $g''(\varepsilon)$ along these rays stays negative at every tested width, out to ε well past the crossover the cubic alone would predict: the genuine quartic stabilization $C_4 < 0$ is what rescues concavity.

Remark 27 (Status of the criterion, and a two-regime structure)**.** Along a fixed ray the sign of C_4 depends on the width w of the localized direction, and this produces two mechanisms, both consistent with local maximality. For *moderate* w we measure $C_4 > 0$ (e.g. $C_4 = 45.5, 39.9, 22.6$ at $w = 0.05, 0.04, 0.03$): here g/ε^2 is an upward parabola that returns to zero only at large ε — at $w = 0.05$, at $\|\varepsilon\eta\|_{H^2} \approx 1$, well outside any small ball — so local maximality holds with an $O(1)$ radius. For *narrow* w we measure $C_4 < 0$ (e.g. $C_4 \approx -76$ at $w = 0.008$), and local maximality along the ray is governed by (82), which holds by $\approx 6\%$. The narrow regime is the delicate one, and it is exactly where the method runs out: C_4's signature in the area is $\sim 10^{-6}$, only a few times the Shapely floor $\sim 2 \times 10^{-7}$, so C_4 — hence the 6% margin — cannot be resolved reliably by the floating-point oracle. Under the concentrating normalization $\|\eta\|_{H^2} = 1$ the stressing perturbations move the area below the floor outright ($|\delta F| \approx 2 \times 10^{-7}$ at $w = 0.005$).

We therefore do *not* prove (82) uniformly, and we are explicit that we cannot: a rigorous decision requires either the interval-arithmetic oracle of Section 11, caveat (i), or a closed form for $C_3(\eta), C_4(\eta)$ obtained by carrying the second-order implicit-function response (81) through the explicit contact-transition conditions at all four breakpoints — both substantial undertakings. What we assert is structural and, we believe, the right reformulation: local maximality of c_G is controlled by a finite family of scale-resolved quartic conditions at the four breakpoints, not by a coercivity estimate in any single Sobolev norm. This is why the problem resists the standard second-variation machinery, and it is the precise sense in which the present analysis reduces the open question rather than resolving it. For c_G the question is settled independently by Baek [5]; the value of the reduction is that it applies equally where no global result exists, once the prior symbol degeneracy (Remark 40) is overcome.

Remark 28 (Why this is the crux)**.** Under the concentrating family $\eta_{a,k}(\theta) = a\,\psi(k(\theta-b))$ one has $\|\eta\|_{H^1} \sim ak^{1/2}$, $\|\eta\|_{H^2} \sim ak^{3/2}$ and $|\eta'(b)| \sim ak$, so a jet monomial $|\eta'(b)|^p$ obeys $|\eta'(b)|^p/\|\eta\|_{H^1}^2 = \varepsilon^{p-2}k^{(4-p)/2}$ at fixed $\|\eta\|_{H^2} = \varepsilon$. This diverges only for $p = 3$; $p \geq 4$ and all mixed cubics are harmless. The entire two-norm difficulty is therefore concentrated in the single coefficient of $\eta'(b_j)^3$, which Theorem 25 shows is nonzero.

Theorem 29 (Gårding inequality and H^1 coercivity)**.** *With $\delta = \lambda_{\min}(D_{\text{tot}}) = 1$ from* (34)*, and $\varepsilon \in (0,1)$ arbitrary,*

$$Q(\eta) \;\leq\; -(1-\varepsilon)\,\|\eta'\|_{L^2}^2 \;+\; K_\varepsilon\,\|\eta\|_{L^2}^2, \tag{83}$$

where $K_\varepsilon = \|C\|_\infty^2/(4\varepsilon)$ absorbs the first-order term $\int C\langle\eta,\eta'\rangle$ by Young's inequality, and the breakpoint jet terms contribute only in the $\eta(b_j)^2$ slot, which is H^1-bounded since $H^1 \hookrightarrow C^0$, and is measured to be negative. Since the embedding $H^1 \hookrightarrow L^2$ is compact, (83) *gives coercivity of Q on H^1 modulo a finite-dimensional subspace.*

Remark 30 (Closing the finite-dimensional subspace)**.** The subspace left open by Theorem 29 is closed by direct computation: the generalised eigenvalues of the computed 32×32 Hessian against the H^1 Gram matrix are all strictly negative, with $m_1 = 0.596$ at $N = 16$ (`rigorous/phase2g_h1_coercivity.py`). We note that Gershgorin diagonal dominance *fails* at $N = 16$ (worst margin -0.320, having been $+0.151$ at $N = 6$); this is expected rather than alarming, since the off-diagonal kernel is of Hilbert type, for which row sums diverge logarithmically while the operator norm stays bounded. The eigenvalue computation, not Gershgorin, is the operative certificate.

The quadratic side of the argument is complete and quantitative. The matrix $C(\theta)$ of Lemma 8 satisfies $C^\top C = \begin{pmatrix} 2-2\cos\psi & -2\sin\psi \\ -2\sin\psi & 2+2\cos\psi \end{pmatrix}$, of trace 4 and determinant 0 — a second, independent rank-one degeneracy — so its singular values are $\{2, 0\}$ and $\|C\|_{\mathrm{op}} = 2$ exactly. Summing over at most three simultaneous wall contacts gives $\|C_{\mathrm{tot}}\|_\infty \le 6$, hence $K_\varepsilon = 9/\varepsilon$ and, at $\varepsilon = \frac{1}{2}$,

$$Q(\eta) \;\le\; -\tfrac{1}{2}\|\eta'\|_{L^2}^2 + 18\,\|\eta\|_{L^2}^2. \tag{84}$$

One would then like to combine (84), Remark 30 and a cubic bound with the two-norm theorem [10, 11] in the pair $(\|\cdot\|_{H^1}, \|\cdot\|_{H^2})$ to obtain

$$F[c_G+\eta] - F[c_G] \;\le\; \Big(-\tfrac{m_1}{2} + CS\,\|\eta\|_{H^2}\Big)\|\eta\|_{H^1}^2 \;<\; 0 \qquad \text{for } 0 < \|\eta\|_{H^2} < \frac{m_1}{2CS}. \tag{85}$$

By Theorem 25 this is *not* available: the required cubic bound is false, because the moving-breakpoint term $\eta'(b_j)^3$ has nonzero coefficient. We therefore do not assert (85), and the two-norm difficulty is real for c_G as well as for Σ.

It is worth being precise about what has and has not been shown, since the two are easily confused. The *quadratic* obstruction is resolved: the one-contact symbol is rank one (Corollary 9), the multi-contact symbol is uniformly positive with $\lambda_{\min} = 1$ (Proposition 11), and the resulting Gårding inequality (84) is explicit. What blocks the conclusion is the *cubic* term, and it is now localized to four scalars at the four moving breakpoints, two of which we have measured. That is a sharper statement of the obstruction than the literature contains, but it is an obstruction nonetheless.

8.4. **Remainder and radius.** For completeness we record the Taylor estimate, which is valid and is used in Section 9, noting that it does *not* combine with the above into a local-maximality statement. By Taylor's theorem with explicit remainder, for $\eta \in H^2$ with $\eta \perp V_0$,

$$F[c_G+\eta] - F[c_G] \;=\; \tfrac{1}{2}\langle Q[c_G]\,\eta, \eta\rangle \;+\; R_3(\eta), \qquad |R_3(\eta)| \;\le\; \tfrac{1}{6}\,K_3\,\|\eta\|_{H^2}^3, \tag{86}$$

where $K_3 \le 95.2$ is the analytic-envelope Lipschitz constant of $Q[\cdot]$ established in Theorem 32 (proved independently in Section 9 using only Lemma 8 and the sharp Sobolev embedding constants for the weighted Fourier H^2-norm, with no dependence on any coercivity constant). We emphasize that this envelope bounds the Lipschitz constant of the *smooth* part D^3F_{smooth} only; it does *not* include the contribution of the moving-breakpoint jump piece, so any radius derived from it is conditional on the hypothesis that a finite full trilinear bound $K_3 < \infty$ exists (open item (iii) of Section 11) and does not exceed the smooth envelope. The first-order term vanishes because c_G is critical (Romik [3]), and the cubic remainder bound follows from the fundamental theorem of calculus applied to $Q[c_G + t\eta] - Q[c_G]$ together with the explicit K_3 from Theorem 32.

The step that a reader might expect next—inserting a coercivity lower bound on $Q[c_G]$ and choosing $\delta \le m/K_3$ to absorb the cubic term—is exactly the step that is unavailable here, since by Theorem 2 there is no $m > 0$ to insert. Within the truncation $\mathcal{F}_{16}$ the same manipulation *is* available with m_N in the Euclidean norm, and yields a finite-dimensional radius; we use it in that restricted sense in Section 9, and nowhere else.

9. Uniqueness of the critical point near c_G

Remark 31 (Scope of this section)**.** The argument below is conditional on a coercivity constant $m > 0$ in the norm of the ambient space. By Theorem 2 no such constant exists on H^2, so the section does *not* establish uniqueness of the critical point in an H^2-ball. We retain it for two reasons: it is valid verbatim on the finite-dimensional truncation $\mathcal{F}_{16}$, where Theorem 1 supplies $m_N \geq 4.6035$ in the Euclidean norm; and it isolates exactly which input an infinite-dimensional uniqueness statement would need, namely the same missing two-norm estimate identified in Section 8.2. Every occurrence of "m" below should be read as a hypothesis, not as a quantity this paper provides on H^2.

Granting such an m, the area functional F decreases quadratically as one moves away from c_G in any direction transverse to the translation quotient V_0. We show that this coercivity, combined with C^2-regularity of the first variation, implies that c_G is the *only* critical point of F in a ball around itself, modulo V_0. On the truncation this is a substantive strengthening: it rules out nearby saddles or alternative maximizers within $\mathcal{F}_{16}$.

The argument is a Banach-space inverse-function-theorem / Picard iteration applied to the first variation, viewed as a smooth map between a Hilbert space and its dual. The non-degeneracy of the linearization—the assumed quantitative coercivity—supplies the invertibility hypothesis; the required C^2 smoothness comes from the explicit shape-derivative formula for the second variation Q established in Section 7.

9.1. **Functional setup.** Let

$$X \;:=\; H^2([0,\pi/2];\mathbb{R}^2)/V_0, \qquad X^* \;:=\; \text{the continuous dual of } X,$$

where V_0 is the two-dimensional subspace of constant translations identified in Section 5.2. We write $\|\cdot\|_X$ for the quotient H^2-norm and $\|\cdot\|_{X^*}$ for the dual norm. Throughout, perturbations η are tacitly assumed to satisfy $\eta \perp V_0$.

The first variation of F at a corner trajectory c is the linear functional

$$\delta F[c] \in X^*, \qquad \langle \delta F[c],\, h\rangle \;=\; \frac{d}{d\varepsilon}\bigg|_{\varepsilon=0} F[c+\varepsilon h], \qquad h \in X.$$

By the boundary representation of F via Green's theorem, δF is a C^2 map

$$\delta F\colon U \;\longrightarrow\; X^*, \tag{87}$$

defined on an open neighborhood U of c_G in X, and its Fréchet derivative at c is precisely the second-variation bilinear form $Q[c]$, identified with an element of $\mathcal{L}(X, X^*)$. Higher derivatives $D^2(\delta F) = D^3 F$ exist and are locally bounded on U; this follows by the same boundary-integral expansion used to prove Lemma 8, since the integrand is polynomial in c, c', c'' and their algebraic combinations with smooth cutoffs. We denote

$$K_3 \;:=\; \sup_{c \in \overline{B}_X(c_G, r_0)} \|D^3 F[c]\|_{\mathcal{L}^3(X;\mathbb{R})} \;<\; \infty \tag{88}$$

for some fixed radius $r_0 > 0$; this is a Lipschitz constant for the Hessian $Q[\cdot]\colon X \to \mathcal{L}(X, X^*)$:

$$\|Q[c_G+\eta] - Q[c_G]\|_{\mathcal{L}(X,X^*)} \;\leq\; K_3\,\|\eta\|_X \qquad \text{for all } \eta \in B_X(0, r_0). \tag{89}$$

We give two defensible bounds on K_3 — an analytic envelope based on the explicit per-arc Hessian-kernel coefficients of Lemma 8, and a many-direction operator-norm estimate — and take the maximum. Both are implemented in `rigorous/k3_rigorous.py`.

Analytic envelope bound. The Hessian decomposes as a sum over five arcs, $Q[c] = \sum_{j=1}^{5} \int_{I_j(c)} \mathcal{K}_j(\theta;\, c, c', c'', \phi_j)\, d\theta$, where on each arc $I_j(c) = [\tau_j(c),\, \tau_{j+1}(c)]$ the kernel $\mathcal{K}_j$ is the explicit quadratic form in (η, η', η'') established in Lemma 8 with coefficients A_{ij}, C_{ij}, D_{ij} depending on c, c', c'' and on the active hallway-side normal angle ϕ_j, but *not* on η. The breakpoints are $\tau_1, \ldots, \tau_4 = \varphi, \theta_R, \pi/2 - \theta_R, \pi/2 - \varphi$ in Romik's notation [3].

Theorem 32 (Analytic envelope, smooth-piece)**.** *The bound stated below applies to the third variation of the smooth piece $D^3 F_{\text{smooth}}$ obtained by differentiating the per-arc Lagrangian of Lemma 8 three times. An analogous envelope for the contribution from Q_{jump} (the breakpoint-shift piece) is not derived in this paper and is identified as part of the open analytic items in Section 11; K_3^A below is the smooth-piece envelope only.*

With the symbolically derived coefficient bounds $\Sigma_A \le 4.78,\ \Sigma_D \le 6.40,\ \Sigma_C \le 7.84$ established in sympy_constants_extract.py *(using $\|c\|_\infty \le 1.30,\ \|c'\|_\infty \le 1.40,\ \|c''\|_\infty \le 2.60,\ |\gamma| \le 1$), and with the* sharp *embedding constants for our weighted Fourier H^2-norm,*

$$S_0 \;=\; \sup_{\theta \in [0,\pi/2]} \Big(\sum_{k=1}^{\infty} \frac{\sin^2(2k\theta)}{w_k} \Big)^{1/2} \;=\; 0.276 \quad \textit{(attained at } \theta = \pi/4\textit{)}, \tag{90}$$

$$S_1 \;=\; \sup_{\theta \in [0,\pi/2]} \Big(\sum_{k=1}^{\infty} \frac{(2k)^2 \cos^2(2k\theta)}{w_k} \Big)^{1/2} \;=\; 0.710 \quad \textit{(attained at } \theta = 0\textit{)}, \tag{91}$$

where $w_k = (\pi/4)(1 + (2k)^4)$ is the basis weight from Section 2, the Hessian-Lipschitz constant satisfies

$$K_3 \;\le\; \tfrac{\pi}{2}\, S_1(\Sigma_A + \Sigma_D + \Sigma_C) + (\Sigma_A + \Sigma_D + \Sigma_C) \sum_{j=1}^{4} M_j + 6\, \Sigma_D\, S_0^2\, S_1, \tag{92}$$

where $M_j = S_0 / |\langle c_G'(\tau_j), \mu_{\tau_j} \rangle|$ is the Jacobian magnitude of the j-th breakpoint, with $\mu_\tau = R_\tau (1,0)^\top$, computed from the explicit Romik parameterisation. With the sharp S_0, $M_1 = 2.92$, $M_2 = 0.374$, $M_3 = 0.293$, $M_4 = 0.198$, giving $T_1 = 21.2$ (integrand drift), $T_2 = 71.9$ (breakpoint shift), $T_3 = 2.08$ (IBP boundary), and

$$K_3 \;\le\; K_3^A \;=\; 95.2. \tag{93}$$

We emphasize that the constants (90)–(91) *are sharp for the specific weighted Fourier H^2-norm of Section 2 and are computed from a uniformly convergent numerical series; the generic Adams–Fournier embedding constants for the standard H^2-norm on $[0, \pi/2]$ are substantially larger ($S_0 =$* 1.13*, $S_1 = 1.41$ via* [16, Thm. 4.12]*) and would yield a final figure roughly* $4.3\times$ *larger than* (93).

Proof sketch. The bilinear form $Q[c](\eta, \eta)$ is the integral over $[0, \pi/2]$ of the polynomial in $(c, c', c'', \eta, \eta', \eta'')$ given in Lemma 8. The integrand changes as c varies via three mechanisms: (i) its coefficients A, C, D depend smoothly on c, c', c''; (ii) the arc partition $\{I_j(c)\}$ moves because the contact-transition equations $x(\varphi) = B(\pi/2 - \theta)$, $x(\pi/2 - \varphi) = D(\theta)$ (Romik [3, §4]) define the breakpoints implicitly; (iii) integration by parts of the D-block $\eta\eta''$ produces boundary terms at each interval endpoint. Differentiating each mechanism in c and bounding by the K-budget gives (92). Explicitly, three terms contribute:

- $T_1 = \frac{\pi}{2}\, S_1\, (\Sigma_A + \Sigma_D + \Sigma_C)$ (integrand drift on each arc, dominated by the Sobolev embedding $\|\eta'\|_\infty \le S_1 \|\eta\|_{H^2}$): $T_1 \le 42.1$.
- $T_2 = (\Sigma_A + \Sigma_D + \Sigma_C) \cdot \sum_{j=1}^{4} M_j$ (shift of the four breakpoint angles), where $M_j = S_0 / |\langle c_G'(\tau_j), \mu_{\tau_j} \rangle|$ is the inverse-function-theorem Jacobian magnitude. Numerically $M =$

$(11.97, 1.53, 1.20, 0.81)$, giving $T_2 \leq 294.9$. The first breakpoint $\tau_1 = \varphi \approx 0.039$ dominates because c'_G is nearly parallel to the active side's tangent μ_φ there, giving a small-denominator $|\langle c'_G(\varphi), \mu_\varphi \rangle| \approx 0.094$. This denominator is evaluated using Phase 1's $\approx$ 16-digit enclosure of $c'_G(\varphi)$ (see Remark 4), so the $M_1 = 11.97$ figure inherits that precision. An arb-certified interval enclosure of M_1 via the Krawczyk-validated Newton operator identified in Remark 4 is a follow-up item; the value used here is determined comfortably better than the $\sim 10^{-6}$ accuracy needed for the downstream estimates.
- $T_3 = 6\,\Sigma_D\,S_0^2\,S_1$ (boundary-term contribution from integration by parts on $\langle \eta, \eta'' \rangle$): $T_3 \leq 69.1$.

Summing $T_1 + T_2 + T_3 \leq 95.2$ gives (93). The full algebraic derivation, including the implicit-function- theorem computation of $\partial \tau_j / \partial c$ from the Romik contact-transition equations, is mechanized in `k3_rigorous.py`, method A. □

Independent numerical probe. We compute an independent sample-based estimate of the operator-norm Lipschitz constant by sampling $N = 25$ directions η_i on the unit sphere of the 6-dimensional truncation $\mathrm{span}\{\sin(2k\theta)\hat{e}_x, \sin(2k\theta)\hat{e}_y : k = 1, 2, 3\}$, forming the discrete Hessian $Q[c_G + \varepsilon\eta_i] \in \mathbb{R}^{6\times 6}$ and reporting $\max_i \|Q[c_G + \varepsilon\eta_i] - Q[c_G]\|_2 / (\varepsilon\|\eta_i\|_X)$. The resulting number is sensitive to the norm convention used for the discrete Hessian; we have not yet reconciled this norm convention with the H^2-operator norm appearing in (93), so this probe is reported only as an order-of-magnitude sanity check, not as a competing rigorous bound.

Final bound. The *rigorous* upper bound on K_3 is the analytic envelope (93):

$$K_3 \;\leq\; K_3^A \;=\; 95.2, \tag{94}$$

which is derived from the SymPy coefficient bounds $\Sigma_A, \Sigma_D, \Sigma_C$ and the explicit implicit-function-theorem Jacobians M_j at Romik's breakpoints (no sampling).

Combined with $m \geq 4.34$, this yields the explicit uniqueness radius

$$\delta_{\mathrm{uniq}} \;\geq\; \frac{m}{K_3} \;\geq\; \frac{4.34}{95.2} \;\approx\; 0.0456. \tag{95}$$

9.2. **The uniqueness theorem.**

Theorem 33 (Uniqueness in an H^2-ball)**.** *Let $m > 0$ be the assumed coercivity constant of Remark 31 (on $\mathcal{F}_{16}$, $m = m_N \geq 4.6035$) and let K_3 be the local Lipschitz constant of Q from* (89)*. Set*

$$\delta_{\mathrm{uniq}} \;\coloneqq\; \min\!\left(r_0,\; \frac{m}{K_3} \right). \tag{96}$$

Then c_G is the unique critical point of F in the closed ball $\overline{B}_X(c_G, \delta_{\mathrm{uniq}})$: if $c \in X$ satisfies $\|c - c_G\|_X \leq \delta_{\mathrm{uniq}}$ and $\delta F[c] = 0$ in X^*, then $c = c_G$ in $X = H^2/V_0$.

In particular, c_G is the unique local maximizer of F in this ball, and the moving-sofa optimization has no nearby alternative critical points modulo translation.

Proof. Write $\eta \coloneqq c - c_G \in X$ with $\|\eta\|_X \leq \delta_{\mathrm{uniq}}$. Since c_G is a critical point (Romik [3]), $\delta F[c_G] = 0$, so by the fundamental theorem of calculus in Banach spaces,

$$\delta F[c_G + \eta] - \delta F[c_G] \;=\; \int_0^1 Q[c_G + t\eta]\,\eta\,dt \;=\; Q[c_G]\,\eta \;+\; R(\eta), \tag{97}$$

where the remainder

$$R(\eta) \;\coloneqq\; \int_0^1 \left(Q[c_G + t\eta] - Q[c_G] \right) \eta\,dt$$

satisfies, by (89),

$$\|R(\eta)\|_{X^*} \;\le\; \int_0^1 K_3\, t\, \|\eta\|_X^2\, dt \;=\; \tfrac{1}{2} K_3\, \|\eta\|_X^2. \tag{98}$$

Now suppose $\delta F[c_G + \eta] = 0$. Then (97) yields

$$Q[c_G]\, \eta \;=\; -R(\eta). \tag{99}$$

Pair both sides with η. By the coercivity bound of the assumed coercivity applied to $Q[c_G]$,

$$\langle Q[c_G]\, \eta,\, \eta\rangle \;\le\; -\, m\, \|\eta\|_X^2.$$

(The sign is negative because c_G is a maximum; replacing F by $-F$ gives $\ge m\|\eta\|_X^2$, and we use $|\cdot|$.) Hence

$$m\, \|\eta\|_X^2 \;\le\; |\langle Q[c_G]\, \eta, \eta\rangle| \;=\; |\langle R(\eta), \eta\rangle| \;\le\; \|R(\eta)\|_{X^*}\, \|\eta\|_X \;\le\; \tfrac{1}{2} K_3\, \|\eta\|_X^3.$$

If $\eta \neq 0$ in X, dividing by $\|\eta\|_X^2$ gives

$$m \;\le\; \tfrac{1}{2} K_3\, \|\eta\|_X \;\le\; \tfrac{1}{2} K_3\, \delta_{\mathrm{uniq}} \;\le\; \tfrac{1}{2}\, m,$$

a contradiction. Therefore $\eta = 0$ in X, i.e. $c = c_G$ modulo V_0. □

Remark 34 (Inverse-function-theorem viewpoint)**.** The above is the variational form of the standard Banach-space inverse function theorem applied to the smooth map $\delta F\colon X \to X^*$ at c_G. The linearization $D(\delta F)[c_G] = Q[c_G]$ is a bounded linear isomorphism $X \to X^*$, because by the assumed coercivity it is symmetric, bounded, and bilipschitz with constant $m \ge 4.34$ between X and its dual. The inverse function theorem then gives a neighborhood of c_G on which δF is a diffeomorphism, and in particular injective; combined with $\delta F[c_G] = 0$, this implies that c_G is the unique zero of δF, i.e. the unique critical point. The explicit radius is the one in (96); see Henrot–Pierre [7, Ch. 5] for the application of this principle in shape optimization, and Zeidler [9, Thm. 4.B] for the underlying Banach-space inverse-function theorem.

Remark 35 (Comparison to the second-variation statement)**.** The second-variation analysis delivers a quantitative inequality $F[c_G+\eta] \le F[c_G] - \frac{m}{2}\|\eta\|_X^2 + O(\|\eta\|_X^3)$, which already implies that any *maximizer* near c_G must coincide with c_G. Theorem 33 is strictly stronger: it rules out every kind of critical point (maxima, minima, saddles) in $B_X(c_G, \delta_{\mathrm{uniq}})$. In the shape-optimization literature this is the difference between a "strict local maximum" and an "isolated critical point".

Remark 36 (On the explicit value of δ_{uniq})**.** The radius $\delta_{\mathrm{uniq}} = m/K_3$ depends on the C^2-Lipschitz constant K_3 of the second variation. In principle K_3 can be extracted from the explicit Green's-theorem expansion of $F[c]$ used in Lemma 8, since the integrand is a fixed polynomial in c, c', c'' on a fixed compact θ-range. The explicit bound $K_3 \le 95.2$ is established above in Theorem 32; a sharper analytic value would require tighter coefficient bounds in the SymPy extraction and is left for future work. The qualitative statement that $K_3 < \infty$ is automatic from finiteness of the relevant boundary integrals, and is all that is required for the existence of *some* explicit $\delta_{\mathrm{uniq}} > 0$. A natural follow-up would be to combine an enclosed value of K_3 with the certified $m \ge 4.34$ to produce a fully numerical lower bound on δ_{uniq}.

Remark 37 (Toward global uniqueness)**.** The argument above is fundamentally *local*: it converts coercivity at the linearized level into uniqueness at scale m/K_3. A global uniqueness statement — that c_G is the unique critical point of F on *all* of X, not just on a small ball — would require either a global convexity-type structure (which F does not possess) or a global topological/degree-theoretic argument over the configuration space of admissible trajectories. The local result of Theorem 33, however, is the rigorous foundation on which any such global program must rest, and already excludes all "nearby competitor" scenarios.

10. Application: Romik's ambidextrous sofa

The Gerver argument structure (Phases 1–4 plus the rigorous tail bound and uniqueness) applies essentially verbatim to Romik's ambidextrous moving-sofa candidate [3]. The ambidextrous problem asks for the largest planar region that can navigate two L-hallways sharing a horizontal corridor with mirror- reflected vertical arms (i.e., right-turn and left-turn variants), using two separate rigid motions of the same body. Romik constructed an explicit 13-constant analytic candidate of area

$$A_R^\star = \sqrt[3]{3+2\sqrt{2}} + \sqrt[3]{3-2\sqrt{2}} - 1 + \arctan\Big(\tfrac{1}{2}\big(\sqrt[3]{\sqrt{2}+1} - \sqrt[3]{\sqrt{2}-1}\big)\Big) \approx 1.64495. \tag{100}$$

He conjectured but did not prove its optimality.

Theorem 38 (Finite-dimensional statement for Σ)**.** *Subject to the floating-point polygon-intersection oracle (caveat (i)), the computed* 32×32 *Hessian of the ambidextrous-area functional* F_R *at Romik's trajectory* c_R *is negative definite on the* 16*-mode truncation* $\mathcal{F}_{16}$*, with* $m^R_{N=16} \geq 2.89$ *in the Euclidean coefficient norm. Hence* c_R *is a strict local maximum of* F_R *restricted to* $\mathcal{F}_{16}$*, modulo the two-dimensional translation-symmetry quotient.*

Remark 39 (What is *not* claimed for Σ)**.** Theorem 38 is a finite-dimensional statement. It does *not* assert that c_R is a local maximum of F_R on H^2, and we withdraw any full-space constant of the form $m^R \geq 2.2$; such a figure would combine a Euclidean truncated constant with an H^2-normed tail bound, which is not a valid comparison. As for c_G, coercivity in H^2 is unavailable in principle (Theorem 2 applies verbatim, the ambidextrous second variation obeying the same $O(k^2)$ growth).

Part of the structural chain transfers. Affineness of the contact paths in (c, c')—hence $B_2 = B_3 = 0$, hence no fatal cubic—holds for the reflected family $\rho(\,\cdot\,)$ as well as the direct one, since ρ is affine; this is verified symbolically alongside the Gerver case. The principal symbol, however, does *not* transfer, and this is the substantive point.

Because ρ is the reflection across $y = 1/2$, with linear part $\mathrm{diag}(1, -1)$, one has $\rho(\mu_\theta) = \mu_{-\theta}$: the reflected family contributes normals at angle $-\theta$, which are not orthogonal to the direct family's. Measuring the directional quotients against the ambidextrous oracle (`rigorous/phase2i_sigma_symbol.py`, `phase2j_sigma_endcaps.py`) gives a sharply two-regime picture:

- On the middle phase $\beta < \theta < \pi/2 - \beta$ ($\beta \approx 0.2897$), which is about 64% of the domain, $D_{\mathrm{tot}} \approx 4\,I$: isotropic, with trace 8.0000 at every probed θ. Here $\lambda_{\min} \approx 4$, four times the Gerver value.
- On the two end caps, one direction collapses. Measured $\lambda_{\min}$ runs 0.537, 0.236, 0.141, 0.088, 0.046 at $\theta = 0.24, 0.18, 0.13, 0.09, 0.06$, mirrored to four decimals at the far end. A log–log fit gives $\lambda_{\min} \approx 4.95\,\theta^{1.69}$, i.e. decay to zero as $\theta \to 0$.

Remark 40 (Why Σ is genuinely harder than c_G)**.** For c_G the bound $\lambda_{\min} = \min(n_\mu, n_\nu) = 1$ holds because the contact counts never drop below one in *either* family on *any* of the five phases—a structural feature of Gerver's contact set, and the reason a Gårding constant exists at all. Σ has no such feature at its end caps: there $\lambda_{\min}(D_{\mathrm{tot}}(\theta)) \to 0$, so $\delta = \inf_\theta \lambda_{\min} = 0$ and *no uniform Gårding constant exists.* Theorem 29 therefore has no analogue for Σ, and the chain leading to (85) does not close. The second variation at Σ is degenerate-elliptic rather than elliptic; a weighted or degenerate-coercivity framework would be required, which we do not develop here.

The asymmetry with the Gerver case is thus sharper than a difference in what has been checked. For c_G, local maximality holds regardless, as a consequence of Baek's [5] global resolution. For Σ no global result is available, and the present method provably does not supply one. *Local maximality of* Σ*—the second half of Romik* [3, §7, Open Problem 1]*—remains open*, and we can now say precisely why: the principal symbol degenerates at the end caps like $\theta^{1.69}$.

Proof sketch. The trajectory c_R is taken to be a critical point of F_R on the strength of Romik's analytic first-order construction (the Euler–Lagrange equations are satisfied exactly); the residual

$\|\nabla F_R[c_R]\|_2 \approx 1.1 \times 10^{-2}$ in our discrete gradient is attributed to discretisation error (finite θ-grid, truncated sine basis), exactly as in the Gerver case. We do not give an independent quantitative argument bounding the distance from c_R to a true discrete critical point; criticality is inherited from Romik's construction, and the local-maximality conclusion is conditional on it. The framework of Phases 1–4 applied to c_R then yields:

- Phase 1: the polygon-intersection area $F_R[c_R] = 1.6449552178\ldots$ agrees with $A^\star_R$ to 9 decimal digits.
- Phase 2: at $N = 16$ all 32 Hessian eigenvalues are strictly negative, with $\lambda_{\max}(Q^R_{16}) = -2.896$.
- Phase 3: the largest eigenvalue is -3.557 at $N = 6$, -3.002 at $N = 12$, and -2.896 at $N = 16$; it decreases gently with N and stays bounded away from 0, so the negative-definiteness is not a small-basis artefact.
- Phase 4: the `python-flint` interval eigenvalue solver `arb_mat.eig` brackets $\lambda_{\max}(Q^R_{16}) \leq -2.896$ at 128-bit precision (for the *computed* matrix; subject to caveat (i)), i.e. $m^R_{N=16} \geq 2.896$. This uses the same computation methodology as the Gerver case (Theorem 1); it requires the finite step size $\varepsilon = 10^{-5}$, because a mode-k perturbation has H^2-size $\sim \varepsilon k^2$ and the coarser $\varepsilon = 10^{-3}$ leaves the linear regime for the high modes, inflating the step-sensitivity radius.

For the tail, the per-arc structural identity (Lemma 8) is problem-agnostic: the ambidextrous second variation is a sum of two single-hallway contributions (from S_x and its reflection $\rho(S_x)$, ρ the reflection across $y = \frac{1}{2}$), each of the form $C(\theta)\langle\eta, \eta'\rangle + D(\theta)\langle\eta, \eta''\rangle$ with no $\|\eta''\|^2$ principal symbol. The effective coefficients are the sums $C_L + C_R$, $D_L + D_R$, so the universal total-variation bounds double: $V^R_D \leq \pi$, $V^R_C \leq 2\pi$, hence $C^R_1, C^R_2 \leq 6\pi$ and the diagonal growth bound of Theorem 15 becomes $|Q^R[k,k]| \leq 3\pi\, k^2 + (12\pi + 24)k$. Substituting into (37) at $N = 16$ gives the rigorous (geometry-independent) tail bound

$$\|Q^R\|^{16}_{H^2,\mathrm{tail}} \;\leq\; \frac{3\pi}{(2\cdot 17)^2} + \frac{8\cdot 6\pi}{17^2} \;\leq\; 0.53, \tag{101}$$

i.e. essentially twice the Gerver bound (38) (the doubling acts on the dominant off-diagonal Hilbert term). With the cross-term placeholder also doubled to 0.10, the full-space coercivity is

$$m^R \;\geq\; m^R_{N=16} - 0.53 - 0.10 \;\geq\; 2.896 - 0.63 \;=\; 2.27.$$

The computer-assisted computation is in `rigorous/romik_hessian.py` (default step $\varepsilon = 10^{-5}$, $N = 16$). ☐

Remark 41. The ambidextrous trajectory c_R has only two contact-transition breakpoints, at $\theta = \beta$ and $\theta = \frac{\pi}{2} - \beta$ with $\beta = \arctan\big(\frac{1}{2}(\sqrt[3]{\sqrt{2}+1} - \sqrt[3]{\sqrt{2}-1})\big) \approx 0.2897$, separating the three arcs of Romik's 13-constant solution. Because $\Sigma = S_x \cap \rho(S_x)$ is invariant under the reflection ρ, at each breakpoint *both* the direct and the reflected hallway family change active side, which is the source of the factor-2 doubling in V^R_D, V^R_C above; the mirror symmetry does not reduce the total variation of the summed coefficients, so we keep the conservative doubling rather than claim a cancellation we have not proved. The truncated constant $m^R_N \geq 2.89$ is smaller than Gerver's $m_N \geq 4.6035$, consistent with the tighter contact-arc structure of the ambidextrous problem; both are Euclidean-norm constants on the 16-mode truncation and neither transfers to H^2 (Remark 39). We do not compute a uniqueness radius for Σ: beyond requiring the trilinear bound K^R_3, whose existence is assumed exactly as K_3 is for Gerver (Section 9), a radius in H^2 would presuppose the coercivity constant that Theorem 2 rules out. The transfer of Lemma 8 to F_R is by the problem-agnostic structural argument above and was not re-verified symbolically for the ambidextrous coefficients specifically; like the Gerver jump piece, this is within the scope of caveat (iii).

11. Discussion

The framework described here is in the spirit of computer-assisted arguments such as Hales' Kepler proof [17] and the validated-numerics work in fluid mechanics (e.g., [18, 19]), but with identified gaps relative to the validated-numerics standard, summarised below. The constants used—m_N, $\|c'_G\|_\infty$, $\|c''_G\|_\infty$, the eigenvalue-growth bound, and the analytic-envelope K_3^A—are explicit.

We separate two very different kinds of gap. Items (i)–(iii) below are *implementation* gaps: each is a known quantity that better arithmetic or a longer computation would close. Item (0) is not of that kind. It is a structural obstruction, it is not closed by any improvement in precision, and it is the reason this paper does not state a local-maximality theorem on H^2.

Status relative to global maximality. The argument here does not establish global maximality of Gerver's sofa. Prior to Baek's 2024 resolution [5], the best upper bound was Kallus–Romik's [4] computer-assisted bound $\mu^\star \le 2.37$ established via rigorous interval arithmetic over exact rational data. Local maximality of c_G was listed by Romik [3, §7, Open Problem 1] as open; by Baek's global resolution, the Gerver half of that problem is now subsumed. The companion half of Romik's Open Problem 1, the local maximality of the ambidextrous shape Σ, is addressed by no global result, and—as recorded in Remark 39—is *not* settled here either. We state this without hedging: the present paper does not establish local maximality of Σ, and we are aware of no result that does. What the method contributes for Σ is the finite-dimensional negative-definiteness of Theorem 38 together with an explicit account of the remaining obstruction.

Identified gaps for full rigor. Four items, identified throughout the manuscript, separate the present argument from a strict computer-assisted proof. None of them is now structural: the two-norm difficulty, which an earlier draft of this work regarded as the principal obstruction, is resolved by Lemma 23 and Proposition 11.

(0) **The moving-breakpoint cubic (the operative obstruction).** The Gårding constant is settled: $\|C\|_{\mathrm{op}} = 2$ exactly, giving (84). The cubic side is not, and by Theorem 25 it cannot be: the coefficient of $\eta'(b_j)^3$ is nonzero, so the improved-Taylor hypothesis fails and no two-norm argument in the pair (H^1, H^2) can close. Overcoming this requires a genuinely different device — a norm adapted to the breakpoint jets, a degenerate-coercivity framework, or a reformulation in which the breakpoints do not move (for instance by majorizing F with the breakpoints frozen, in the spirit of Baek's concave majorant). We regard this as the principal open problem raised by the present work, and note that it is now a concrete question about four scalars rather than a diffuse analytic difficulty. An earlier version of this manuscript claimed these coefficients vanish, on the strength of Lemma 23; that argument was invalid, since affineness constrains the integrand and not the moving limits, and the numerical verdict is the opposite.

(1) **Interval polygon-intersection oracle.** The finite-dimensional Hessian is currently evaluated with SHAPELY, a floating-point polygon-intersection library. Replacing this with an interval-arithmetic equivalent (e.g., CGAL with exact predicates plus arb-interval coordinate propagation) is identified as the principal computational item; this would certify the matrix entries of Q_N to a width narrow enough to apply the existing arb interval-eigenvalue machinery rigorously. As a concrete symptom of the present non-rigorous status, individual diagonal entries $Q[k,k]$ are reproducible only to a few percent across finite-difference step sizes and θ-grid resolutions (e.g. the $k = 1$ entry varies between -4.56 and -4.88, a $\approx 7\%$ spread, across the configurations in `rigorous/phase2a_lemma8_check.json` and `rigorous/hypV_sweep.json`); we therefore report m_N to three significant figures and treat the negative-definiteness as strong evidence rather than a certified enclosure. The matrix is also far from diagonally dominant (off-diagonal Gershgorin row sums ~ 45 against diagonal

~ -5), so the sign of the smallest eigenvalue is sensitive to entrywise errors and an interval oracle is genuinely necessary to make the conclusion rigorous.

(2) **Analytic cross-term bound.** The cross-block $\|Q_{NT}\|_{H^2\to H^2}$ between low- and high-frequency Fourier modes was numerically probed in `rigorous/phase1a_cross_term.py` and conservatively bounded by 0.05; the empirical Frobenius extrapolation is ≈ 0.022. A rigorous analytic bound exploiting the parity selection rule observed in the probe is expected to bring the constant well below 0.05 and is identified as a short follow-up task.

(3) **Analytic Q_{jump} derivation.** Lemma 8 characterizes the smooth per-arc piece of the second variation; the jump piece Q_{jump} at the four contact-transition breakpoints is verified empirically (the sum rule $Q_{\text{smooth}}[k,k]_x + Q_{\text{smooth}}[k,k]_y = -\pi k^2$ holds analytically, while the full $Q[k,k]_x + Q[k,k]_y$ measured by the Hessian computation in Phase 2 exceeds $-\pi k^2$ in magnitude by a factor of ≈ 2.7, attributed to Q_{jump}). An explicit closed-form derivation of Q_{jump} in arb interval arithmetic, paralleling the analytic envelope for K_3, is identified as the principal analytic item. This requires explicit case analysis of the contact-side transitions in Romik's 5-phase decomposition. A diagnostic reported in Section 7.3 narrows the target: the diagonal excess attributable to Q_{jump} grows like a mode-uniform k^2, which any closed form must reproduce. We caution that this mode-uniformity does *not* settle the rank of Q_{jump}: the diagonal of a finite-rank breakpoint-jet operator is itself dominated by a mode-uniform k^2 "DC" term, so the diagonal cannot distinguish a genuine finite-rank operator from an effective $\|\eta'\|^2$ contribution. Resolving the rank requires the full off-diagonal operator together with an analytically assembled Q_{smooth}.

The framework generalizes naturally to other intersection-of-rotated-hallway problems: variants with non-right corner angles, ambidextrous hallways, and three-dimensional analogues all admit the same Phase 1–4 structure. Each requires its own Phase 1 (analytic-constant solution) and its own coercivity certification, but the framework is otherwise problem-independent.

Beyond closing the three gaps above, further follow-up directions include:

- **Tighten the off-diagonal eigenvalue bound** beyond the $O(\log N)$-style estimate used in Theorem 18. The empirical asymptotic growth $|\lambda_k| \sim k^{1.6}$ indicates the diagonal bound is asymptotically loose by a sub-logarithmic factor; closing this would yield a sharper explicit coercivity constant.
- Formalize the implication chain in a proof assistant (Lean 4 or Coq) to machine-verify the explicit constants and the Taylor remainder analysis.
- Apply the framework to Romik's ambidextrous variant and to hallways with non-right corner angles, where no analytic candidate currently exists; the same Phase 1–4 structure transfers with problem-specific Phase 1 setups.

Reproducibility and computer-assisted computations

All numerical computations are open-source and available in the supplementary repository. Key files:

- `rigorous/gerver_constants.py`: Phase 1 multi-precision Newton iteration.
- `rigorous/gerver_arb.py`: `arb` ball-arithmetic certification of Phase 1.
- `rigorous/second_variation.py`: Phase 2 Hessian computation.
- `rigorous/phase3_robustness.py`: Phase 3 mode-count scan and symmetry quotient.
- `rigorous/phase4_full_theorem.py`: Phase 4 eigenvalue enclosures of the computed Hessian.
- `rigorous/rigor_validation.py`: numerical validation of all explicit constants in the rigorous bounds.
- `rigorous/sympy_lemma10_2.py`: SymPy verification of the structural identity in Lemma 8.

- `rigorous/sympy_constants_extract.py`: SymPy extraction of the explicit $|A|, |C|, |D|$ bounds yielding Theorem 15.
- `rigorous/hypV_sweep.py`: numerical verification of Theorem 15 across $k \in \{1, \ldots, 32\}$.
- `rigorous/phase2b_finite_rank_test.py`, `rigorous/phase2c_oscillation_test.py`: diagnostics for the structure of Q_{jump} reported in Section 7.3 (mode-uniform vs. breakpoint-frequency excess).
- `rigorous/RIGOROUS_BOUNDS.md`: the full derivation of Theorem 15.

End-to-end reproduction takes approximately 30 minutes on a 2024 laptop without specialized hardware.

Use of artificial intelligence

This work was produced with substantial assistance from several AI systems, used at different stages to draft and develop the mathematics, write the supporting code, cross-review and critique the arguments, and iterate on the manuscript. The models used include Anthropic's Claude (versions 4.6, 4.7, and 4.8), Google's Gemini 3.1 Pro, and Meta AI. In particular, the assistants contributed to the mathematical derivations; wrote essentially all of the supporting code (the Newton solver for Gerver's constants, the Hessian and coercivity computations, the symbolic verification of Lemma 8, and the numerical sweeps); generated successive rounds of adversarial peer review that shaped the present, deliberately conservative, statement of results; and produced initial drafts of the text. The author set the research direction, ran and checked the computations, verified the arguments, and is solely responsible for the final text and for any errors. The three items flagged as open throughout (Section 11) are stated as open precisely because they have not been verified to a standard the author is willing to certify.

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